Scientific Presentation (Seminary)

Luiz Henrique Rodrigues

InterSCity Project
Software Sistems Lab
IME - USP

May 10, 2019

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Summary

1. Introduction
2. Speech: The Words You Say
3. Structure: The Strategy You Choose
4. Visual Aids: Your Supporting Cast
5. Delivery: You, the Room, and the Audience
6. Conclusion

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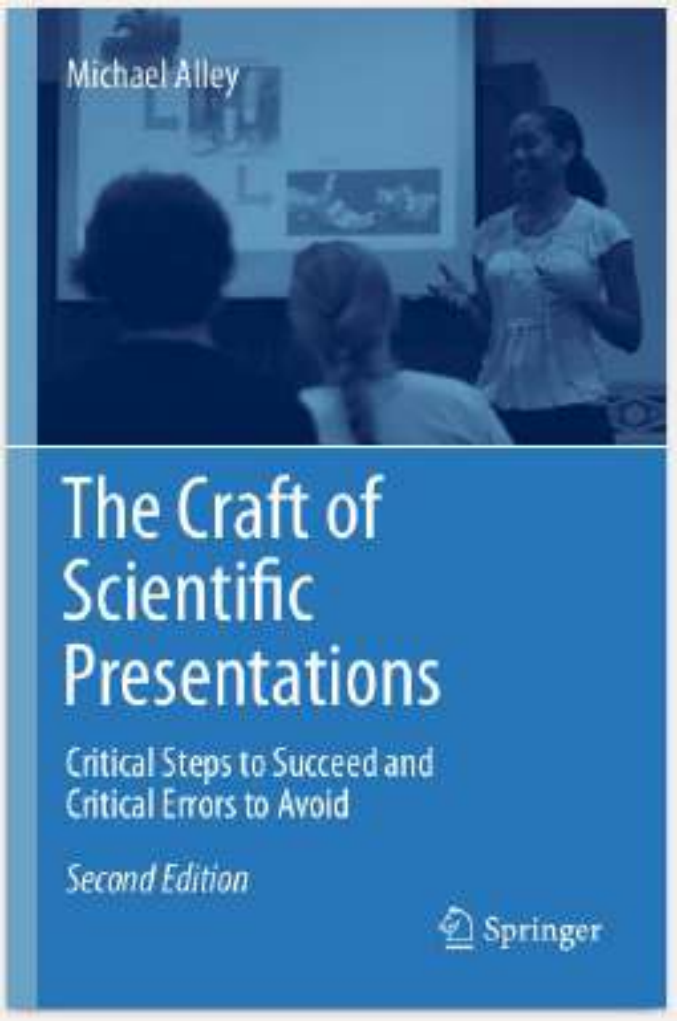
Main Reference



https://link.springer.com/book/10.1007/978-1-4419-8279-7

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Second Reference

Giving an effective presentation:
Using Powerpoint and structuring a scientific talk

based on a presentation at the
2005 Pew Foundation meeting by
Susan McConnell
Department of Biological Sciences
Stanford University

https://player.slideplayer.com/6/5662281/ and
https://profiles.stanford.edu/susan-mcconnell

Before planning your talk think about its purpose, the audience you will be talking to, and the setting

- Do not assume the audience will all be **experts**;
- Never **underestimate** your audience!
- Check on the **time** that has been allotted to you;
- **How big** is the room?

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Excellent scientific presentations are marked by **content**, **passion**, and a **keen sense** of the audience

- **Content** - Stephen Hawking

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Excellent scientific presentations are marked by **content**, **passion**, and a **keen sense** of the audience

- **Passion** - Linus Pauling

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Excellent scientific presentations are marked by **content**, **passion**, and a **keen sense** of the audience

- Achieving the **keen sense** of audience is not easy.

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Speech: The Words You Say

The **matching** of

- what is said to the **audience**,
- **purpose**
- **occasion**.

- Our challenge maintains the **audience's attention**.

After all, if the audience drifts off, then no communication occurs.

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CRITICAL ERROR 1: Giving the Wrong Speech

- In **CRITICAL ERROR 1: Giving the Wrong Speech**, you assess what they know, why they are there, and what biases they hold
- The **CRITICAL ERROR 1: Giving the Wrong Speech** are often a blend of informing and persuading — and sometimes inspiring
- **CRITICAL ERROR 1: Giving the Wrong Speech**, although often overlooked, can greatly affect the way you present

Critical Error 1: Giving the Wrong Speech

- On January 27, 1986,
  Conference call (2 hours - 13 presentation slides) faxed from Morton Thiokol, NASA **remained unconvinced of the danger**.

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Critical Error 1: Giving the Wrong Speech

- On January 27, 1986,
  Conference call (2 hours - 13 presentation slides) faxed from Morton Thiokol, NASA **remained unconvinced of the danger**.
- The next day, 63 seconds into the launch, a solid rocket booster of the space shuttle **Challenger exploded**, killing all seven crew members on board.

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First Slide

Temperature Concern on

SRM Joints

27 Jan 1986

Second Slide

HISTORY OF O-RING DAMAGE ON SRM FIELD JOINTS

		Cross Sectional View			Top View		
	SRM No.	Erosion Depth (in.)	Perimeter Affected (deg)	Nominal Dia. (in.)	Length of Max Erosion (in.)	Total Heat Affected Length (in.)	Clocking Location (deg)
61A LH Center Field**	22A	None	None	0.280	None	None	36° - 66°
61A LH CENTER FIELD**	22A	NONE	NONE	0.280	NONE	NONE	338° - 18°
51C LH Forward Field**	15A	0.010	154.0	0.280	4.25	5.25	163
51C RH Center Field (prim)***	15B	0.038	130.0	0.280	12.50	58.75	354
51C RH Center Field (sec)***	15B	None	45.0	0.280	None	29.50	354
41C RH Forward Field	13B	0.028	110.0	0.280	3.00	None	275
41C LH Aft Field*	11A	None	None	0.280	None	None	–
41C LH Forward Field	10A	0.040	217.0	0.280	3.00	14.50	351
STS-2 RH Aft Field	28	0.053	116.0	0.280	–	–	50

*Hot gas path detected in putty. Indication of heat on O-ring, but no damage.
**Soot behind primary O-ring.
***Soot behind primary O-ring, heat affected secondary O-ring.

Clocking rotation of leak check port - 0 deg.

OTHER SRM-15 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY AND NO SOOT HEAR OR BEYOND THE PRIMARY O-RING

SRM-22 FORWARD FIELD JOINT HAD PUTTY PATH TO PRIMARY O-RING, BUT NO O-RING EROSION AND NO SOOT BLOWBY. OTHER SRM-22 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY.

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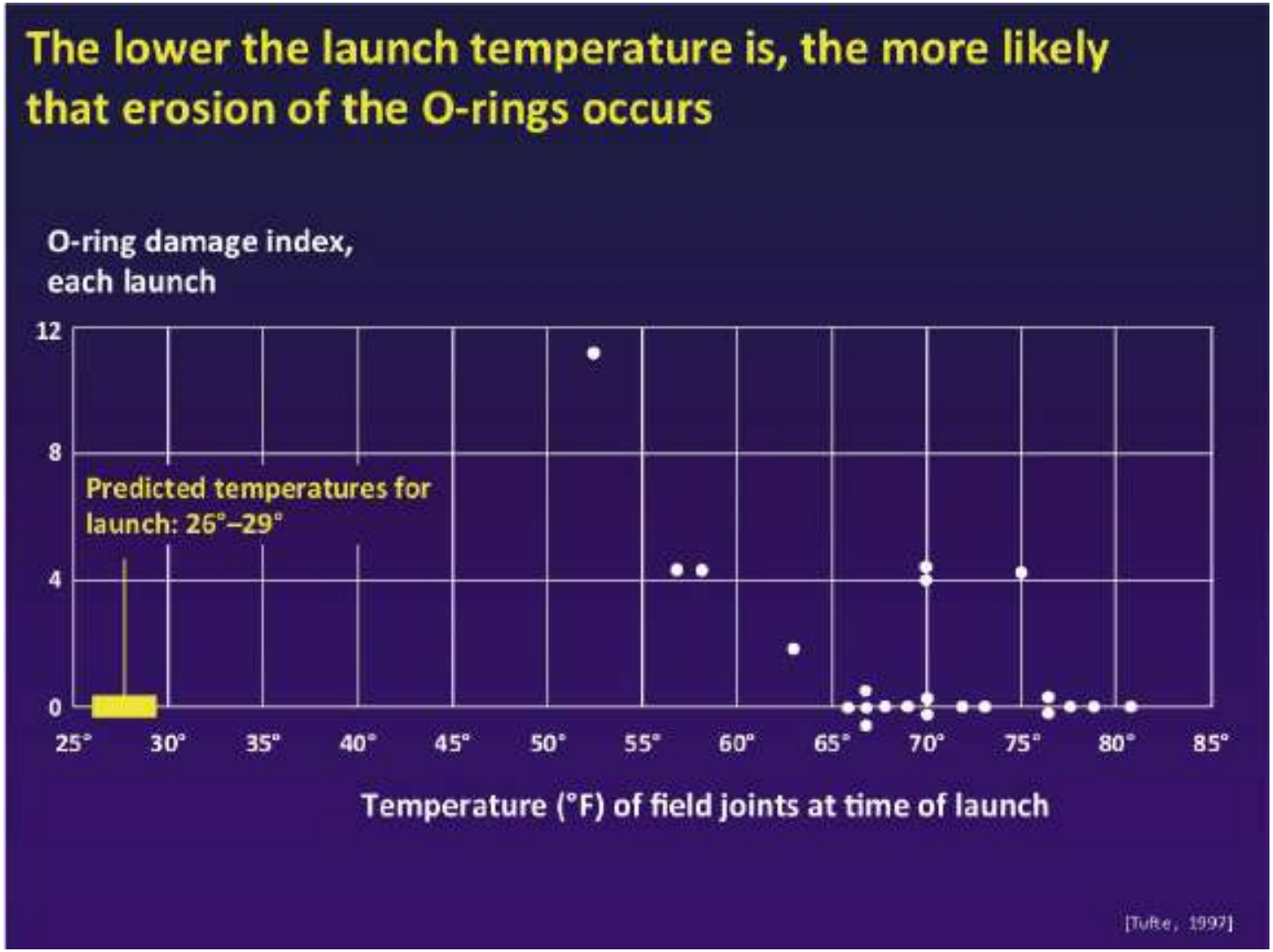
Second Slide **Proposed**



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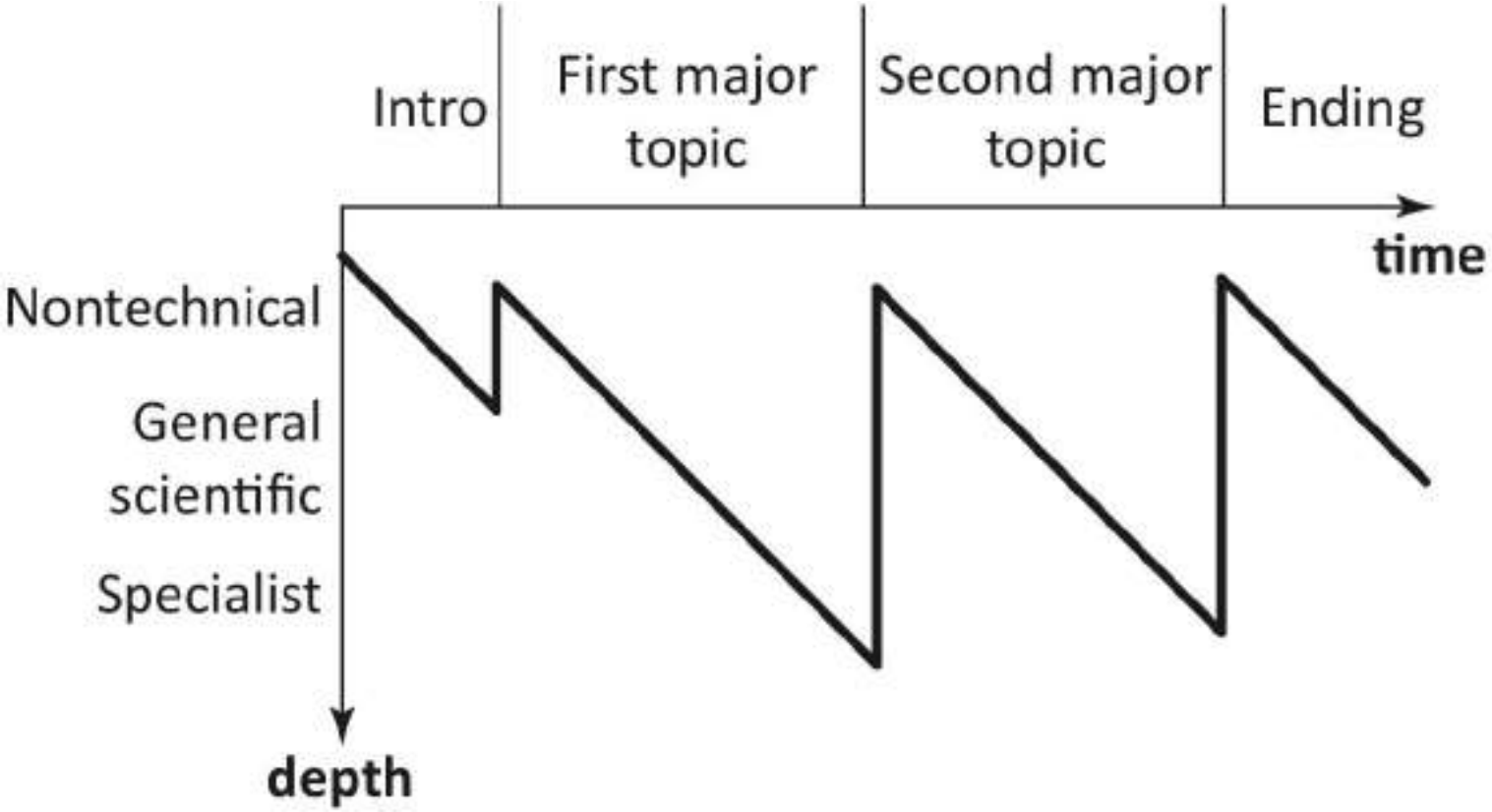
Targeting multiple audiences



Timeline showing presenter reaching multiple audiences

CRITICAL ERROR 2: Boring Your Audience

- In **CRITICAL ERROR 2: Boring Your Audience**, you assess what they know, why they are there, and what biases they hold
- The **CRITICAL ERROR 2: Boring Your Audience** Boring Your Audience are often a blend of informing and persuading — and sometimes inspiring
- **CRITICAL ERROR 2:Boring Your Audience**, Boring Your Audience although often overlooked, can greatly affect the way you present

Critical Error 2: Boring Your Audience

Perhaps the **most stinging criticism** for a presentation is that the talk is boring. Many talks are boring because the speaker projects

bulleted list after bulleted list.

- fsdfds fsdf a s fadsfsa
- sddfds sadf sd fasdf
- asdfgfg ds fads
- fsdfds sdf asf ggda
- sddfds sdf afasdfdafds
- asdfgfg madf adf a faasdf
- sdfo ao fa faofasdfadf
- aflsdl asof saos dfoa ssdf.

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Critical Error 2: Boring Your Audience

Strategies for many of these speakers is the use:

Stories can be engaging and memorable

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Physicists: Story 1

https://elikatakimoto.com/2013/06/08/dialogos/

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Critical Error 2: Boring Your Audience

Strategies for many of these speakers is the use:

- **Stories** can be engaging and memorable
- **Examples and analogies** can help audiences understand unfamiliar concepts
- Making a **personal connection** is a way to connect with the emotions of audiences
- **Humor**, when appropriate, can energize an audience

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Humor

http://assinesalvat.com.br/colecoes/filosofia/

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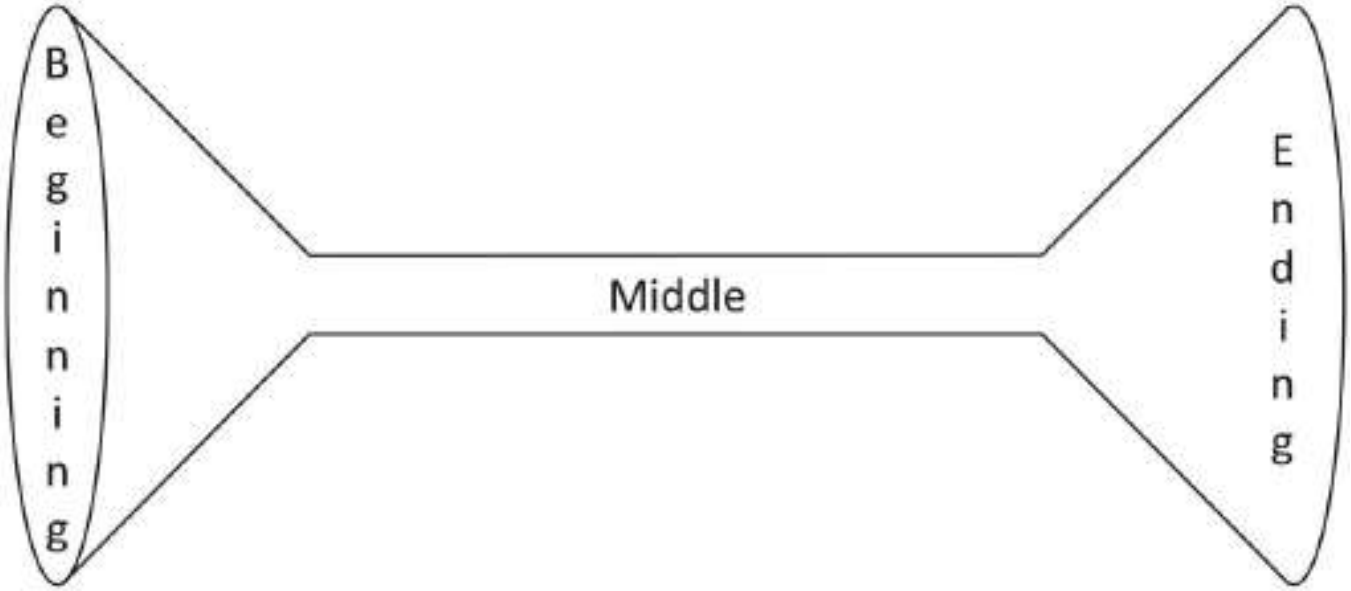
A guide choosing the entry point, keeping the audience on that path (depicted with the dotted line), and highlighting the important sights

Organization is the path up the mountain of your work



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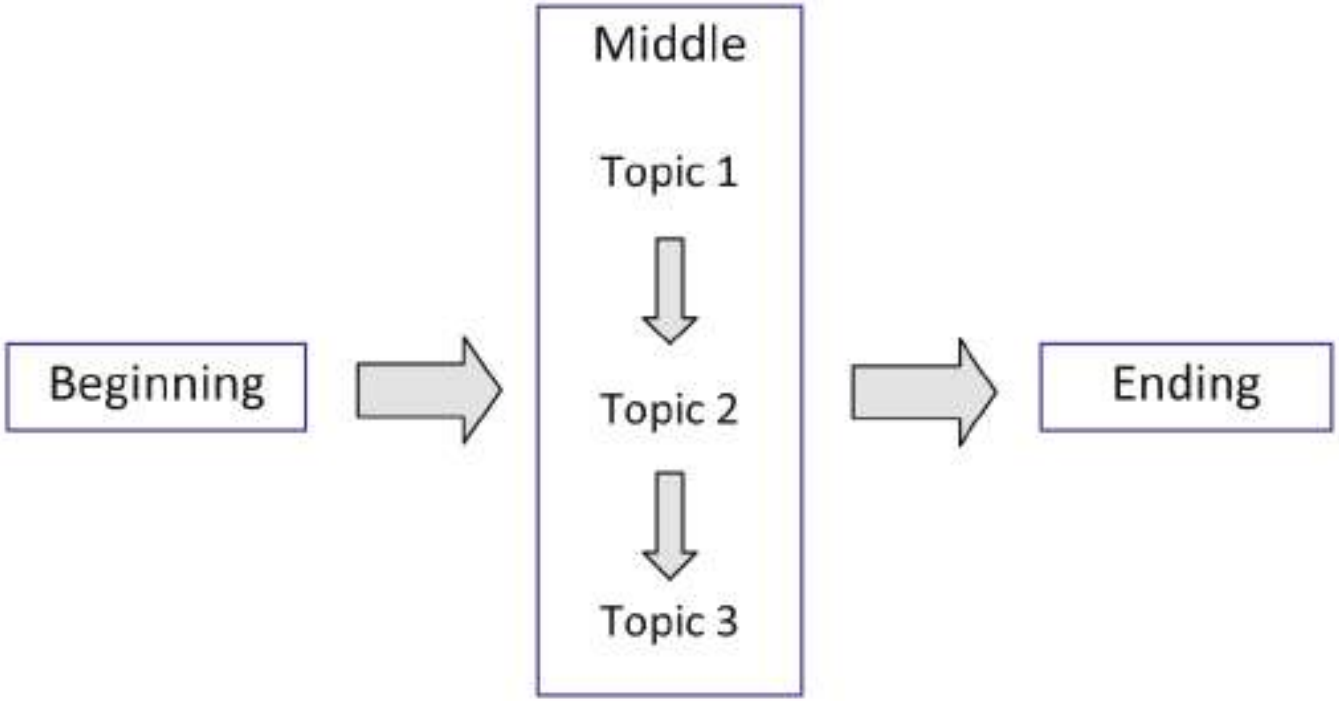
Transitions keep the audience on the trail



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Emphasis tells the audience when to appreciate the view

The structure of a good talk: start broad, get specific, and end broad

Start with the biggest questions and get progressively more specific

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Emphasis tells the audience when to appreciate the view

A powerful tool in a talk is a "home slide"

Design and introduce a "home slide" that you'll come back to at each major transition in your talk.

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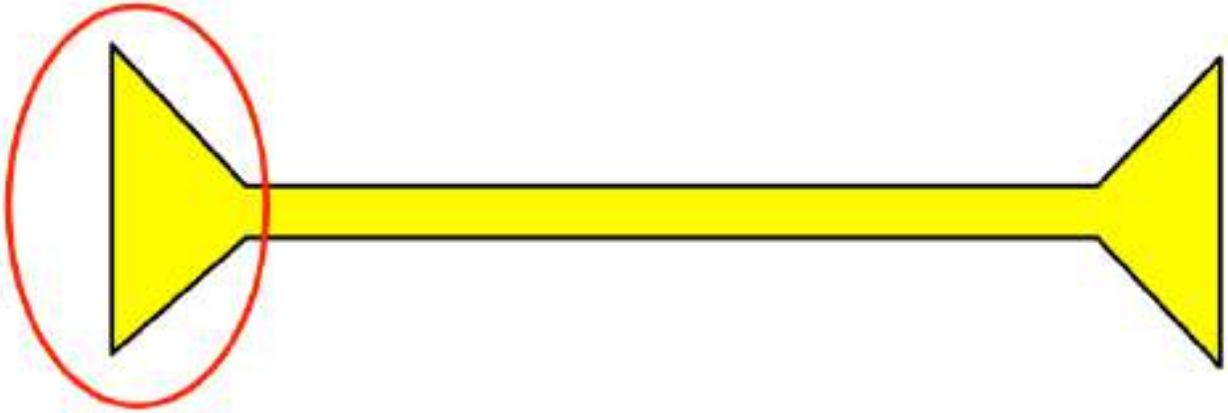
Home Slide

HOME SLIDE

Emphasis tells the audience when to appreciate the view

A powerful tool in a talk is a "home slide"



Now we'll build an introduction and a home slide that puts the previous data into context.

ME-USP

Physicists: Story 2

https://elikatakimoto.com/2013/06/08/dialogos/

Emphasis tells the audience when to appreciate the view

The structure of a good talk: start broad, get specific, and end broad

After going into depth, come back to your home slide to make transitions

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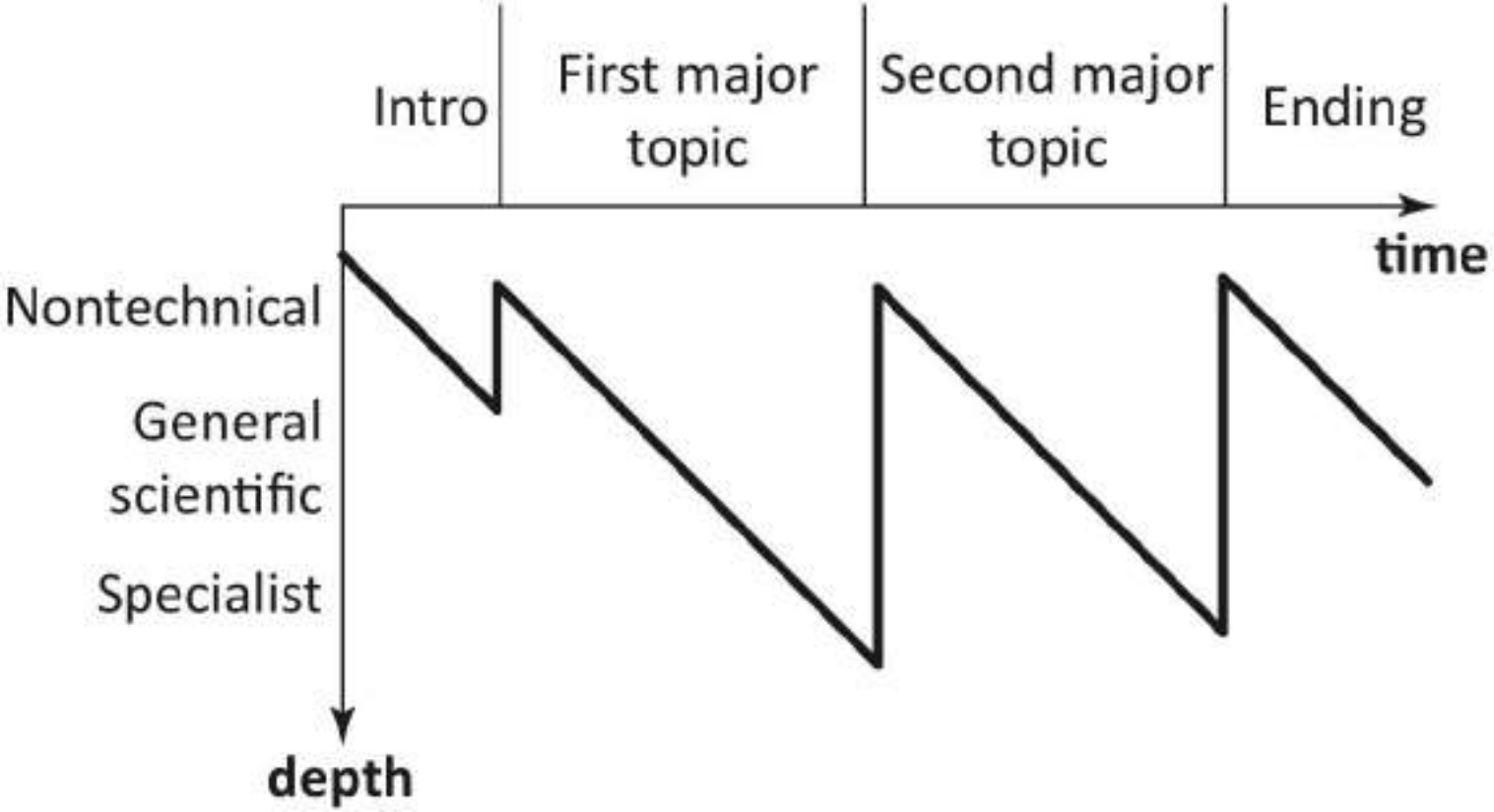
Emphasis tells the audience when to appreciate the view



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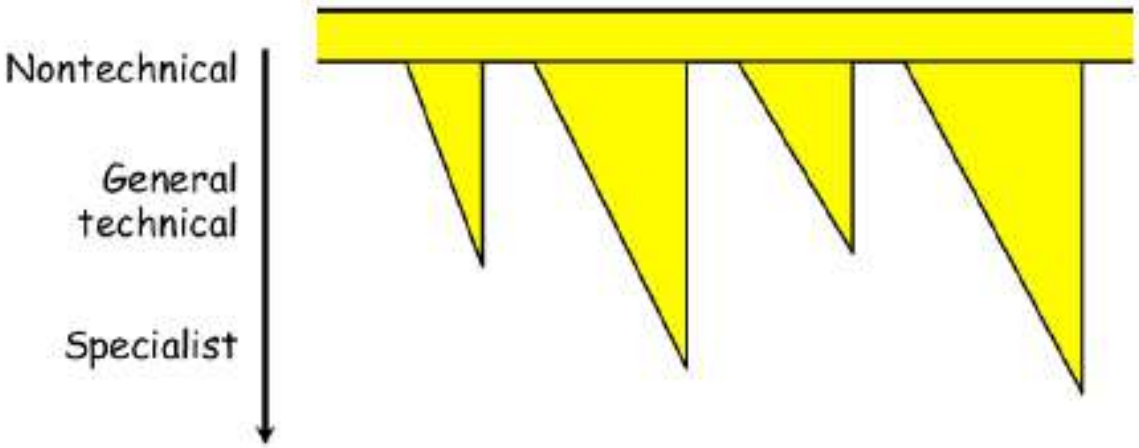
Emphasis tells the audience when to appreciate the view

After going into depth, come back to your home slide to make transitions



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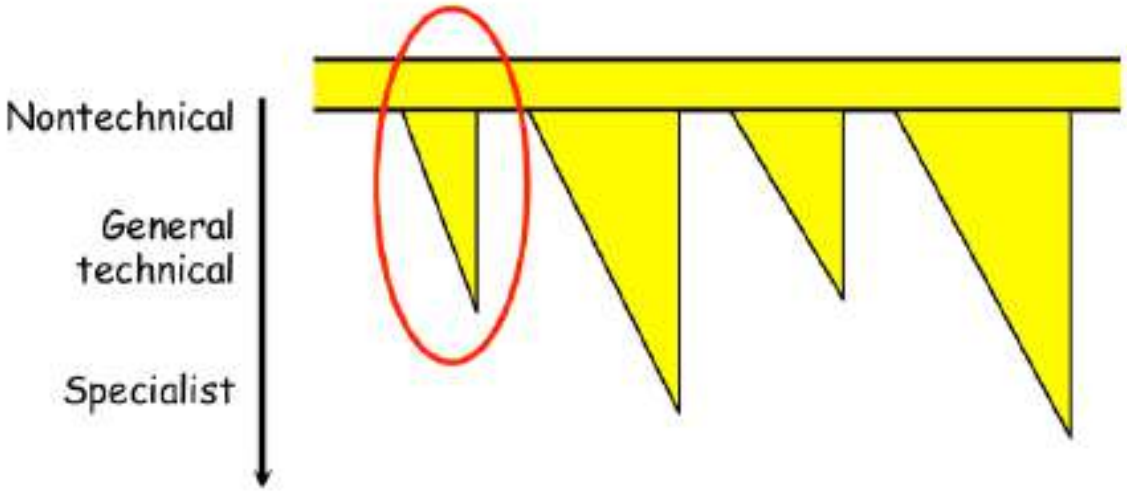
Emphasis tells the audience when to appreciate the view

Let's review "episode 1" (which we've already designed) and add a home slide



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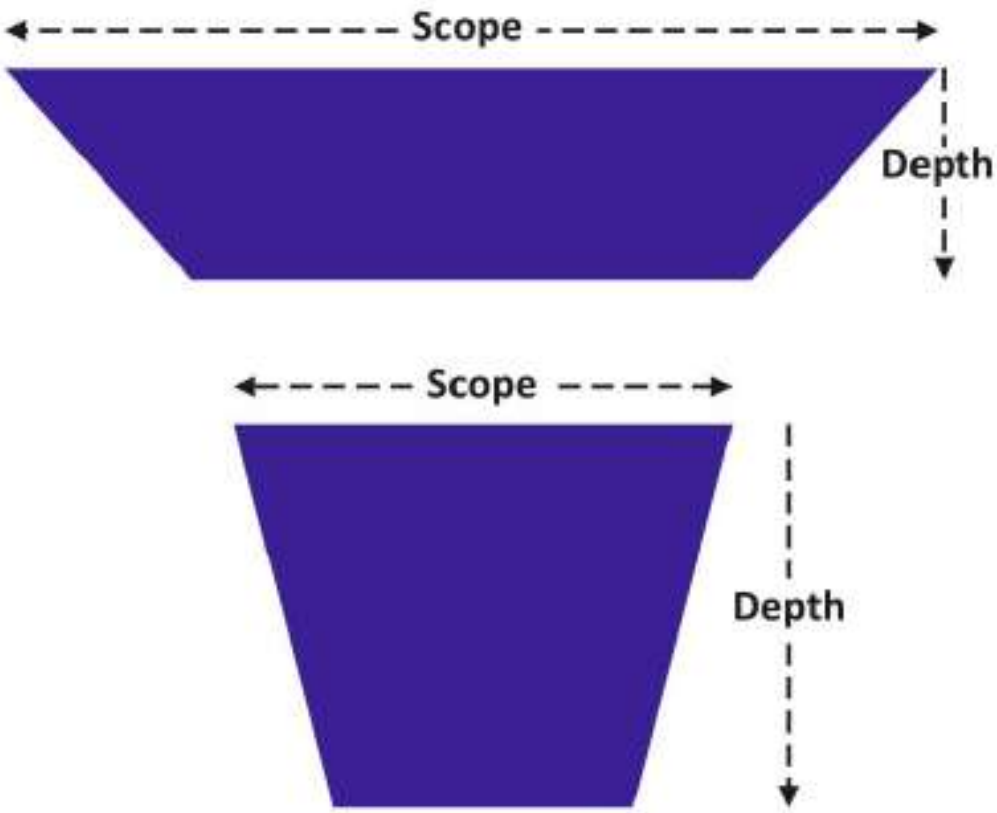
CRITICAL ERROR 3: Trying to Cover Too Much

- In **CRITICAL ERROR 3**, Trying to Cover Too Muchyou assess what they know, why they are there, and what biases they hold
- The **CRITICAL ERROR 3** Trying to Cover Too Much are often a blend of informing and persuading — and sometimes inspiring
- **CRITICAL ERROR 3**, Trying to Cover Too Muchalthough often overlooked, can greatly affect the way you present

Critical Error 3: Trying to Cover Too Much

Relationship of depth and scope.



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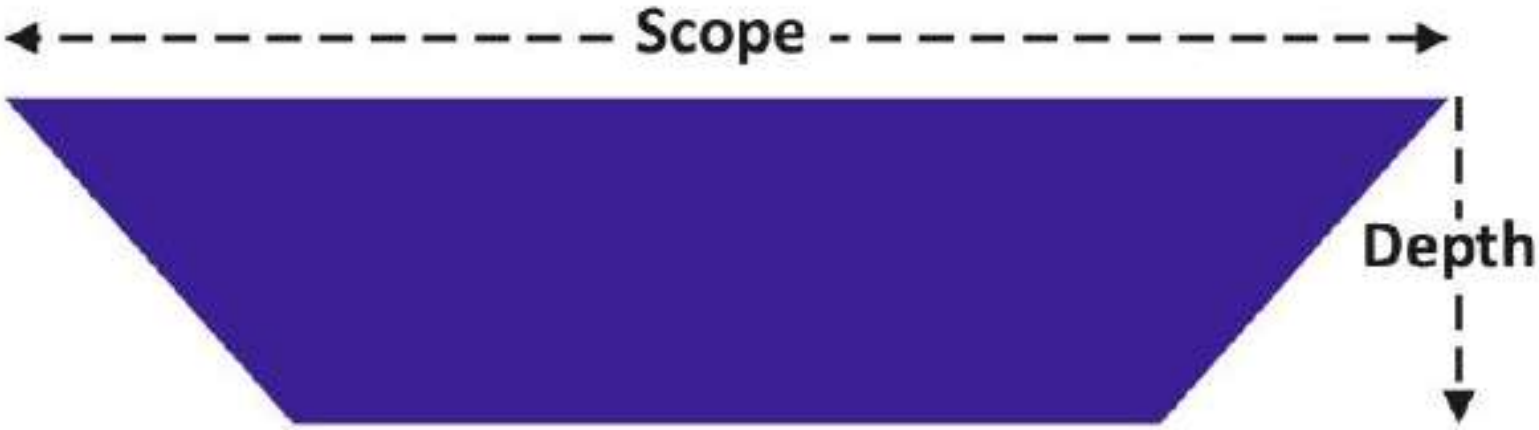
Critical Error 3: Trying to Cover Too Much

- Many talks fail because the scope is **too broad**



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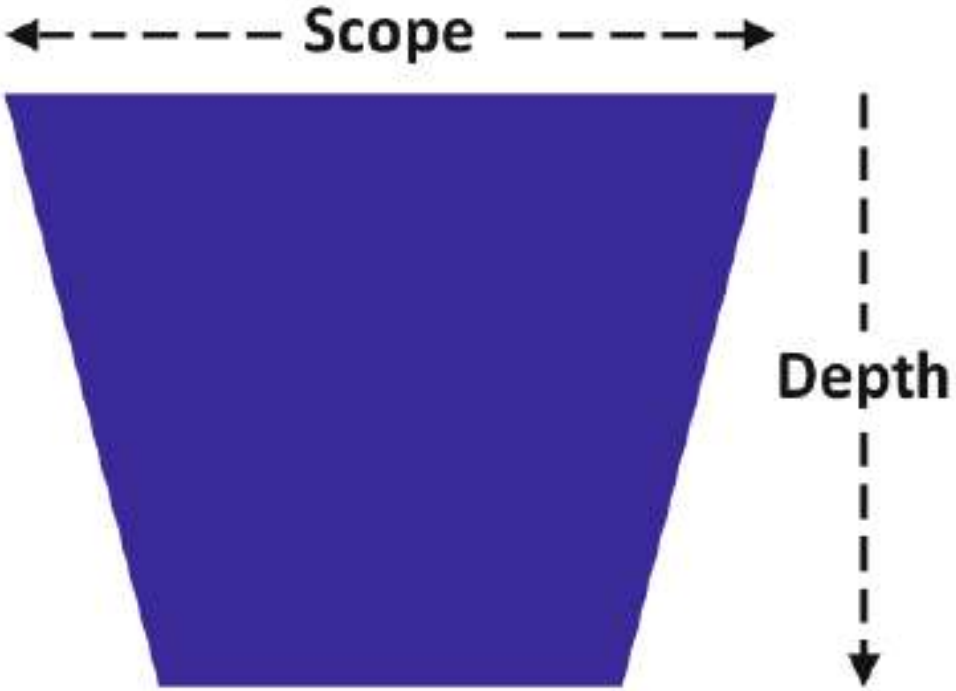
Critical Error 3: Trying to Cover Too Much

- Many talks fail because the scope is **too deep**



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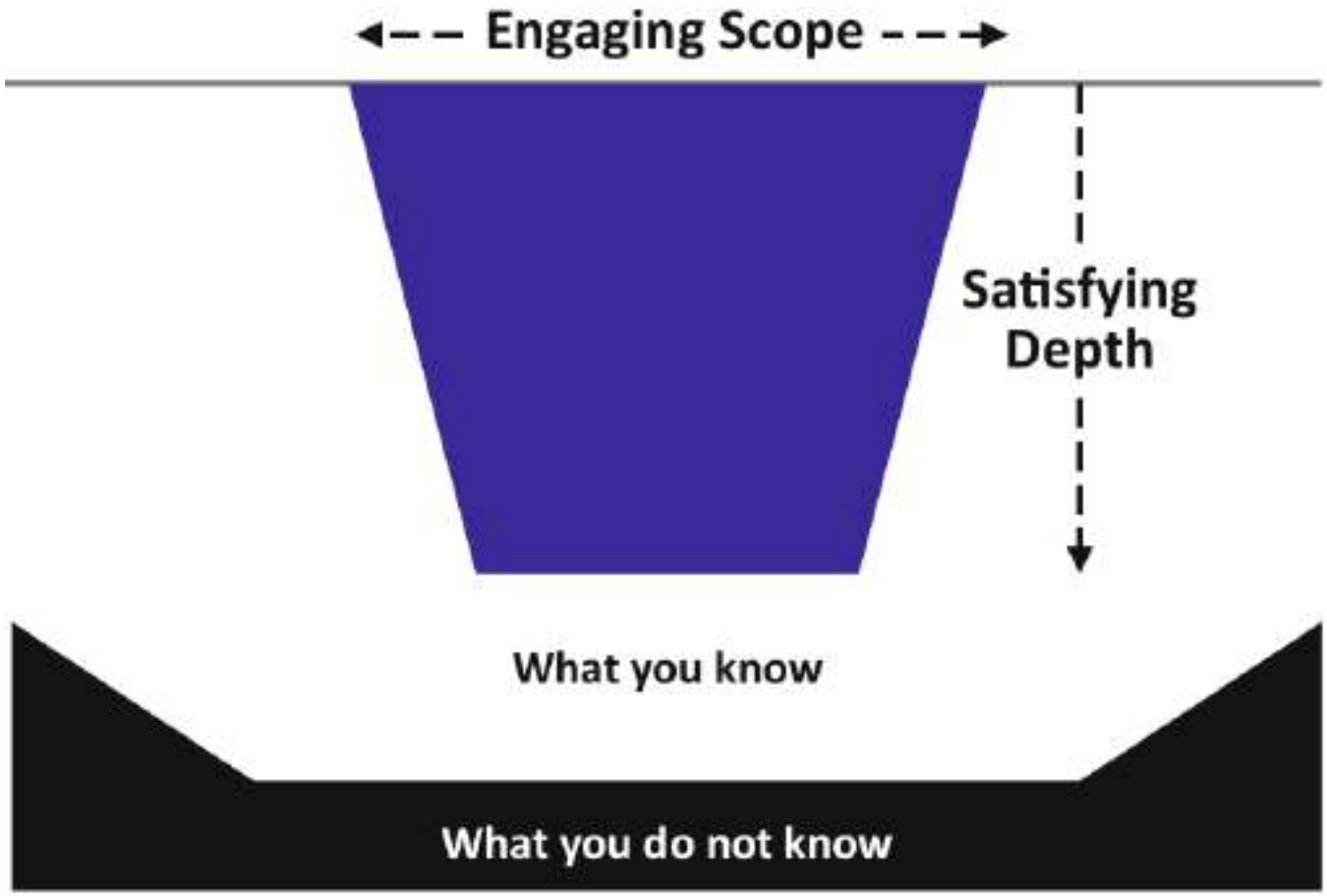
Ideal selection of scope and depth



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Critical Error 4: Losing the Audience from the Start

- In **CRITICAL ERROR 4 : Losing the Audience from the Start**. what they know, why they are there, and what biases they hold
- The **CRITICAL ERROR 4 : Losing the Audience from the Start** Trying to Cover Too Much are often a blend of informing and persuading — and sometimes inspiring
- **CRITICAL ERROR 4**, Trying to Cover Too Muchalthough often overlooked, can greatly affect the way you present

Critical Error 4: Losing the Audience from the Start

The beginning should:

- identify the **boundaries of the subject**
- show **the importance**
- provide needed background and **establish credibility**
- **memorably map** the talk

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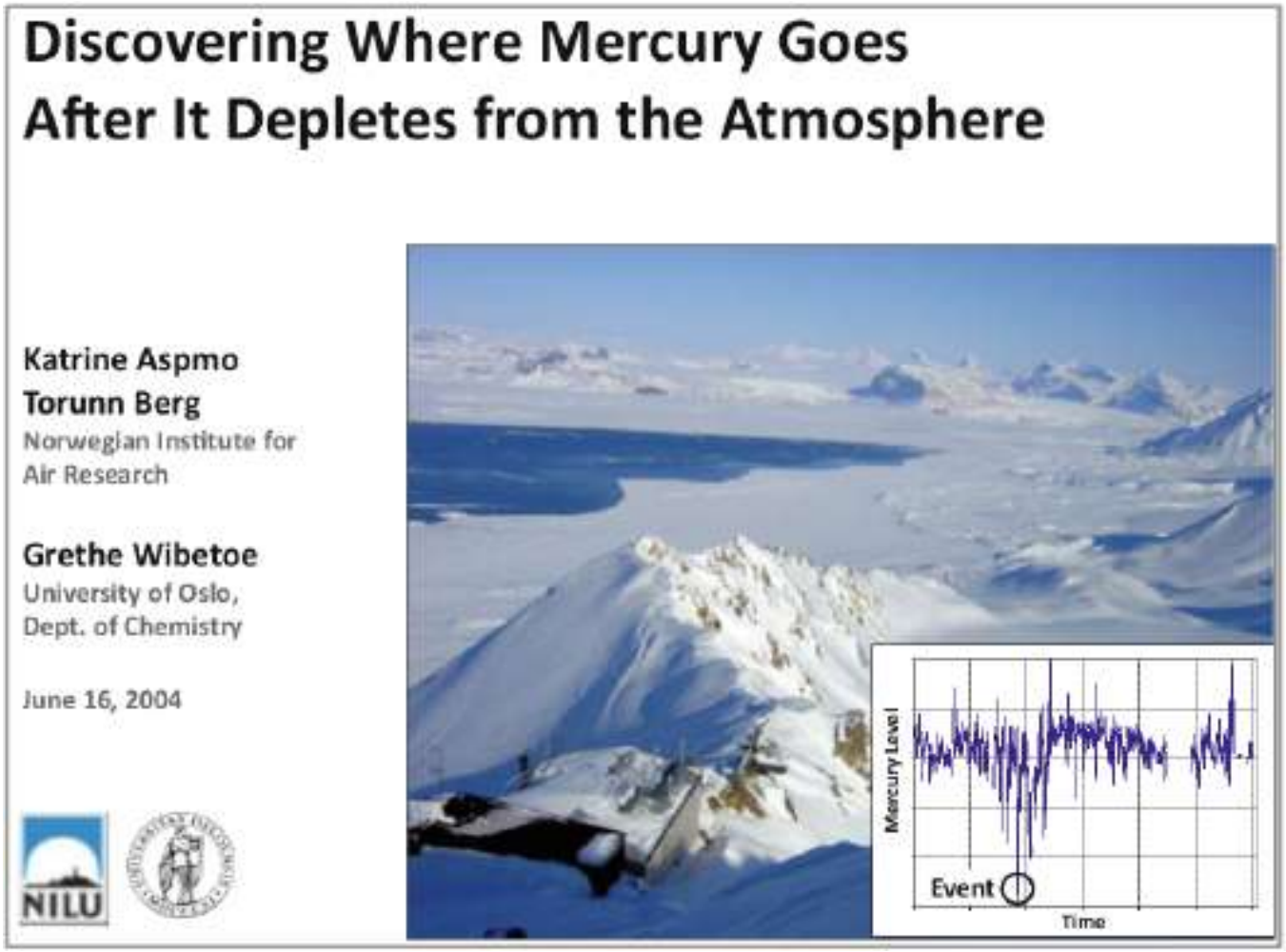
Critical Error 4: Losing the Audience from the Start



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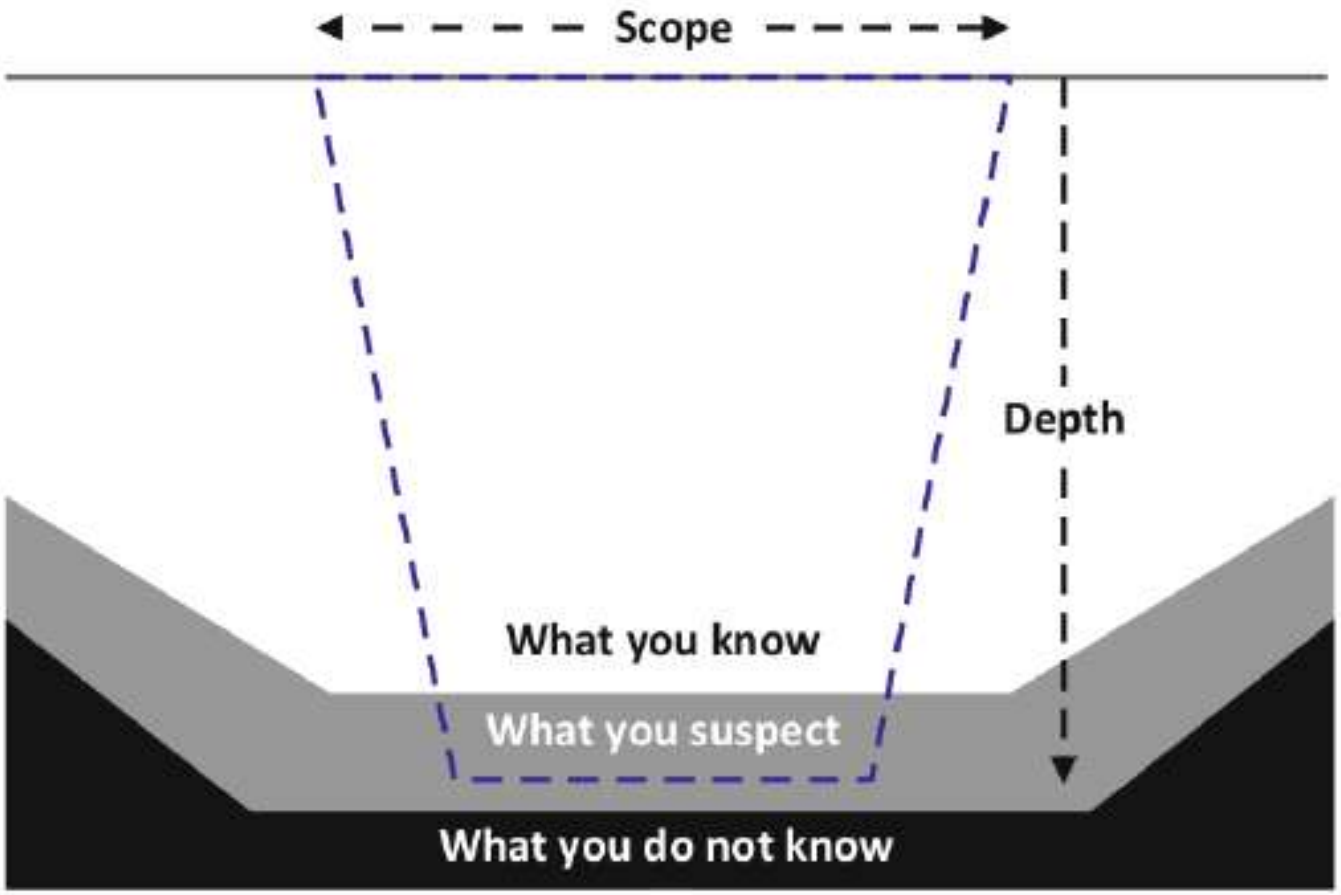
Critical Error 5: Losing the Audience on the Trail

- In **Critical Error 5: Losing the Audience on the Trail**, Trying to Cover Too Muchyou assess what they know, why they are there, and what biases they hold
- The **Critical Error 5: Losing the Audience on the Trail** Trying to Cover Too Much are often a blend of informing and persuading — and sometimes inspiring
- **Critical Error 5: Losing the Audience on the Trail**, Trying to Cover Too Muchalthough often overlooked, can greatly affect the way you

Critical Error 5: Losing the Audience on the Trail

The speaker has to choose a destination that the audience can reach



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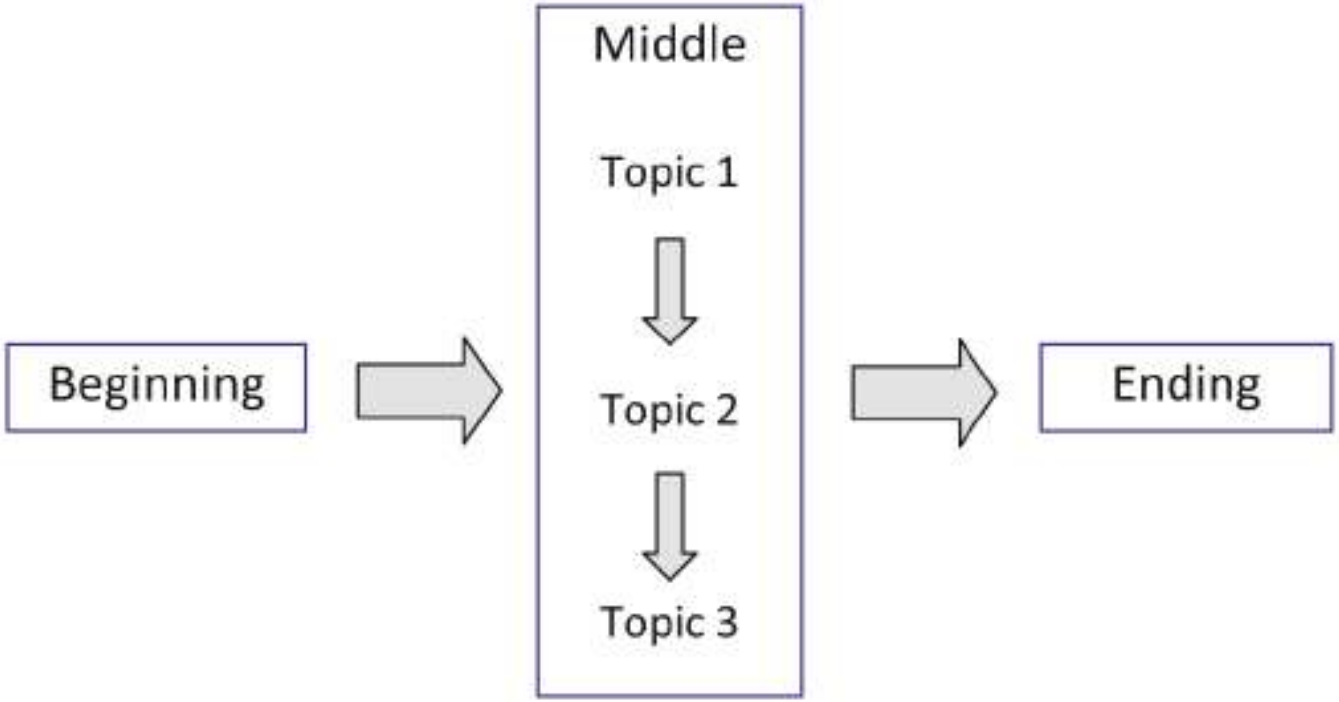
Transitions keep the audience on the trail

The speaker has to signal changes in direction



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Critical Error 5: Losing the Audience on the Trail

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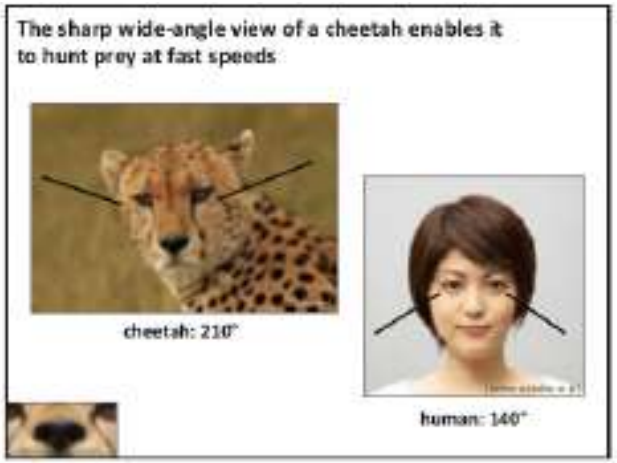

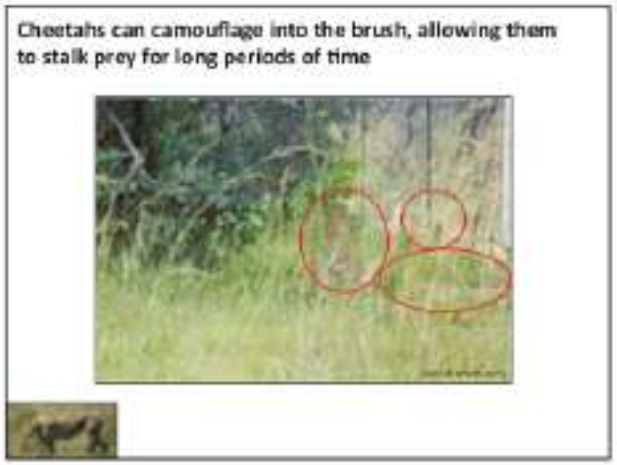

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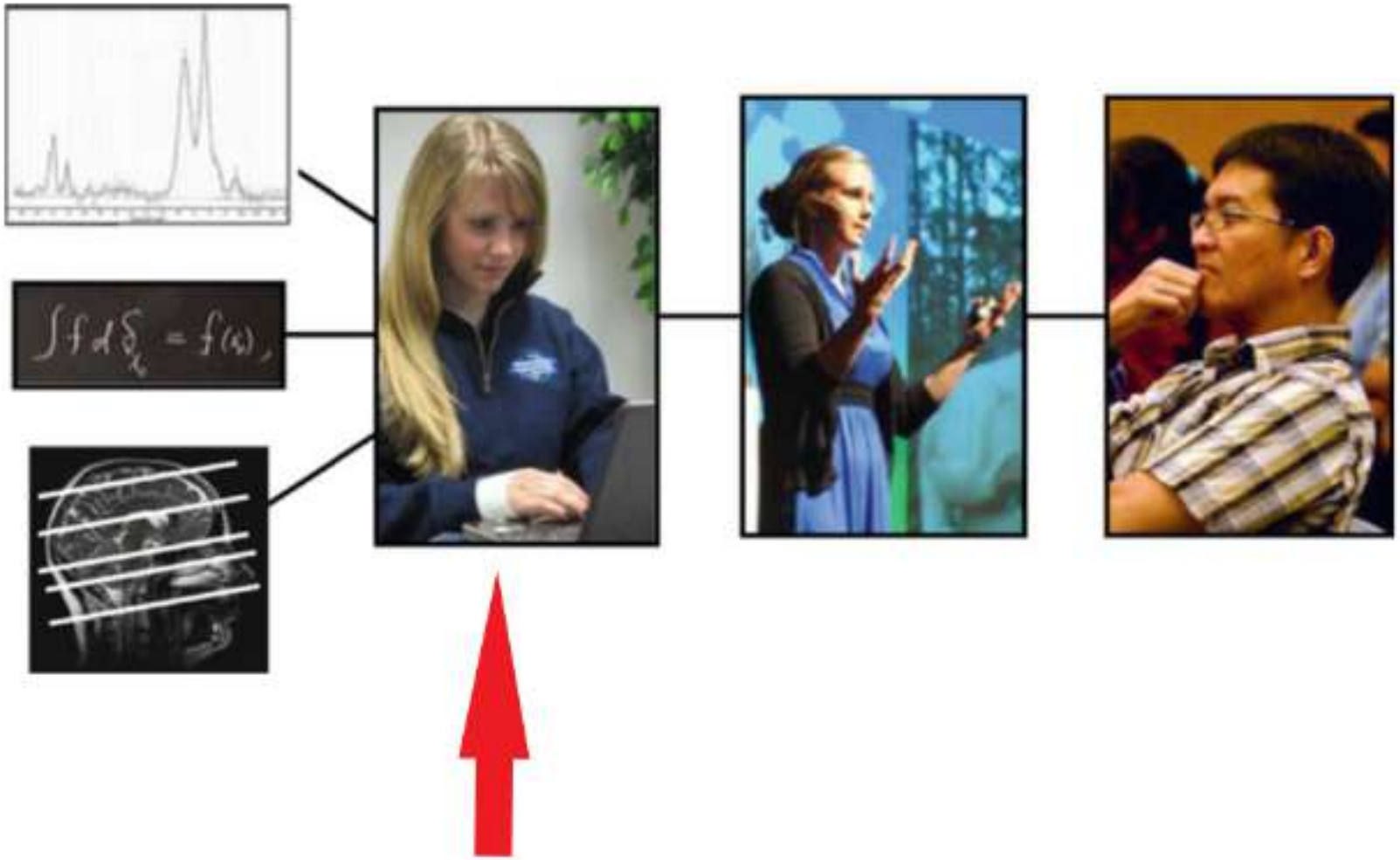
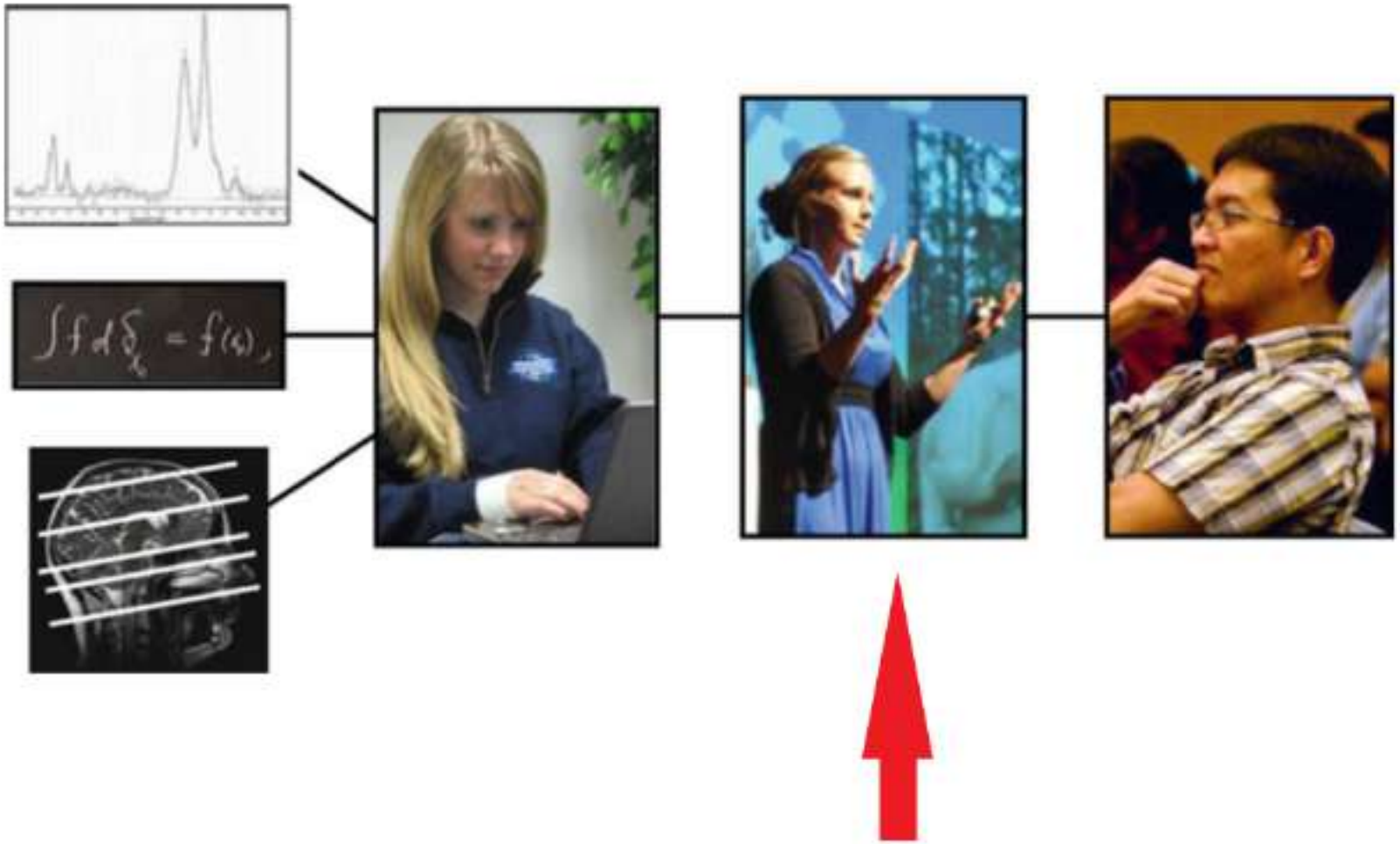


Scientific Presentation (Seminary)

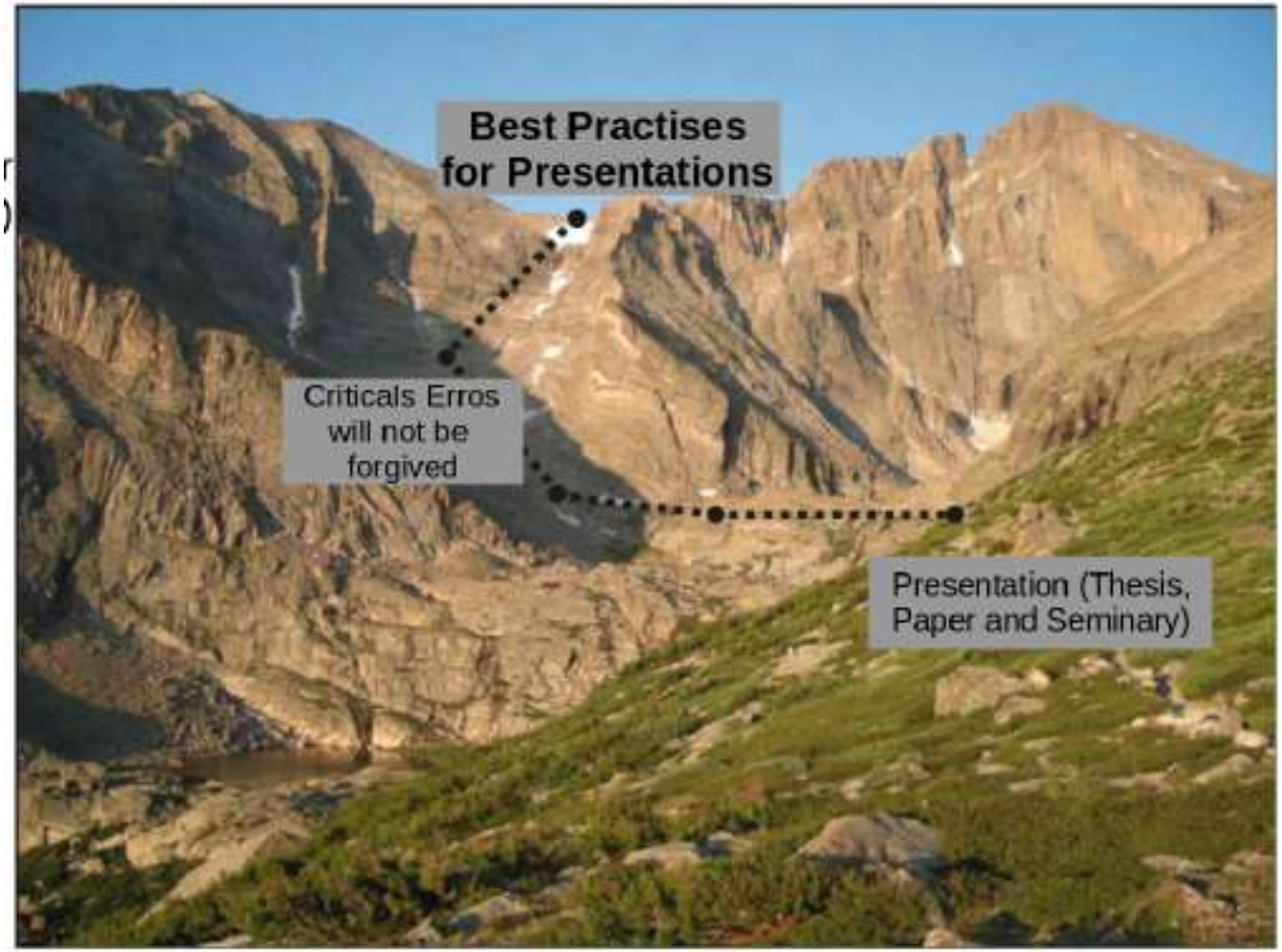


Scientific Presentation (Seminary)

Critical Error 6: Not Anticipating the Audience's Bias

- In **Critical Error 6: Not Anticipating the Audience's Bias**,
- The **Critical Error 6: Not Anticipating the Audience's Bias** Trying to Cover Too Much are often a blend of informing and persuading — and sometimes inspiring
- **Critical Error 6: Not Anticipating the Audience's Bias**, Trying to Cover Too Muchalthough often overlooked, can greatly affect the way you present

Mapping of how sub-assertions can help an audience reach a main assertion in a presentation



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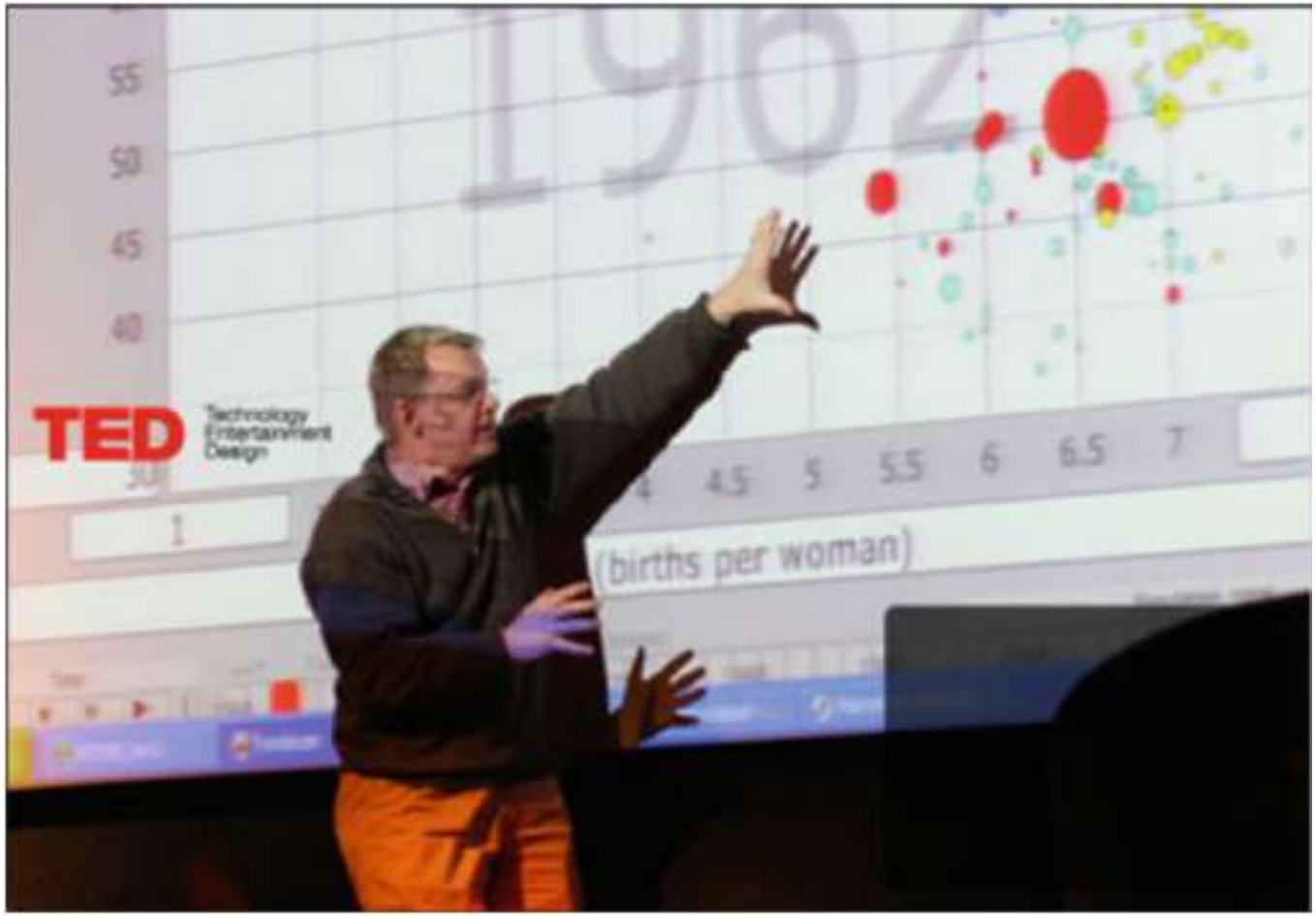
An audience is more likely to believe your argument if they know and appreciate the assertions

Hans Rosling presenting at TED



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An effective argument provides ample evidence for the assertions - **Appeals to Logic**

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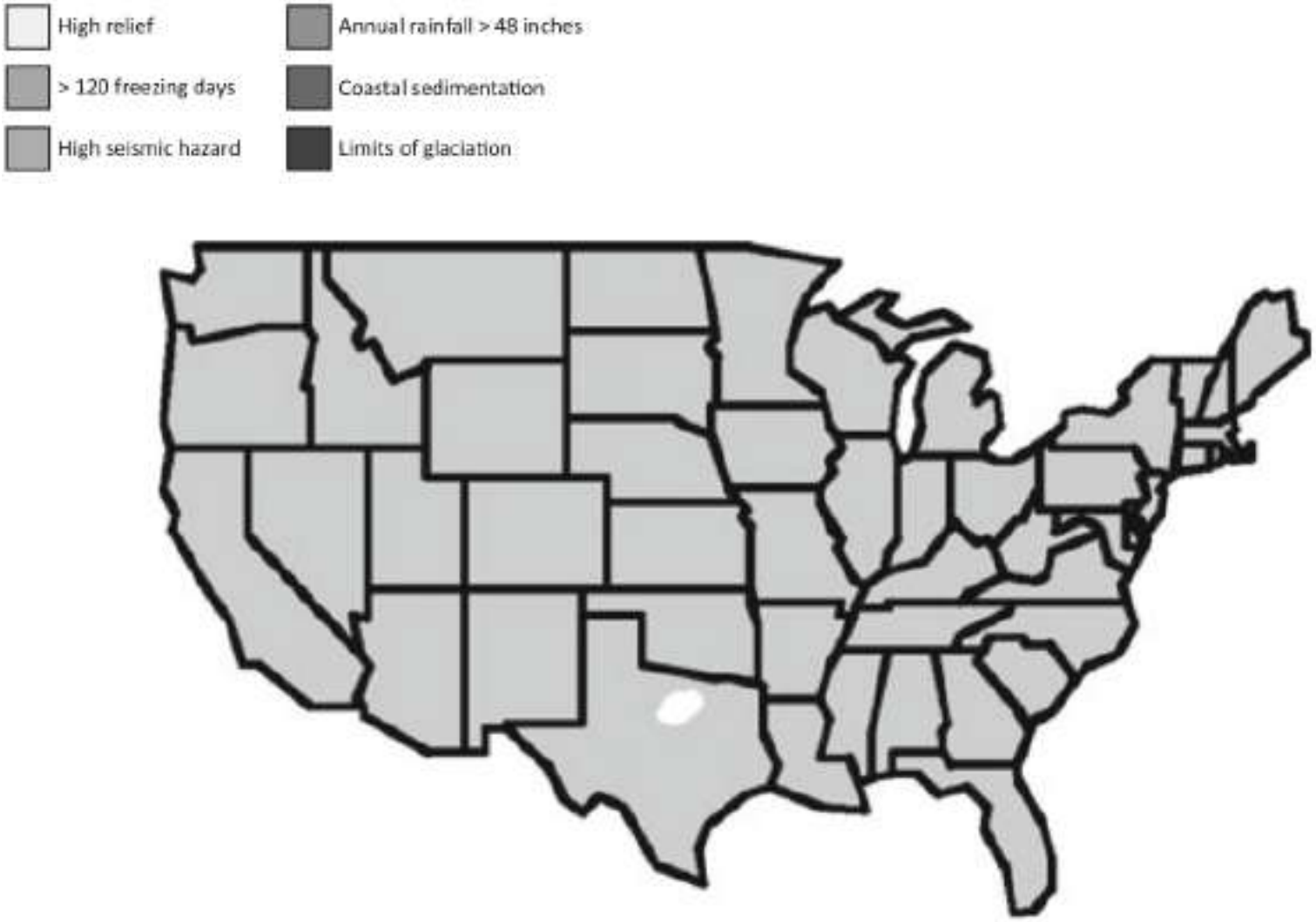
An effective argument provides ample evidence for the assertions - **Appeals to Logic**



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An effective argument provides ample evidence for the assertions - **Appeals to character**

- **Vitamin C**

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An effective argument provides ample evidence for the assertions - **Appeals to character**

- **Vitamin C**

An effective argument provides ample evidence for the assertions -**Appeals to emotion**

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With an antagonistic audience, building **credibility is crucial**

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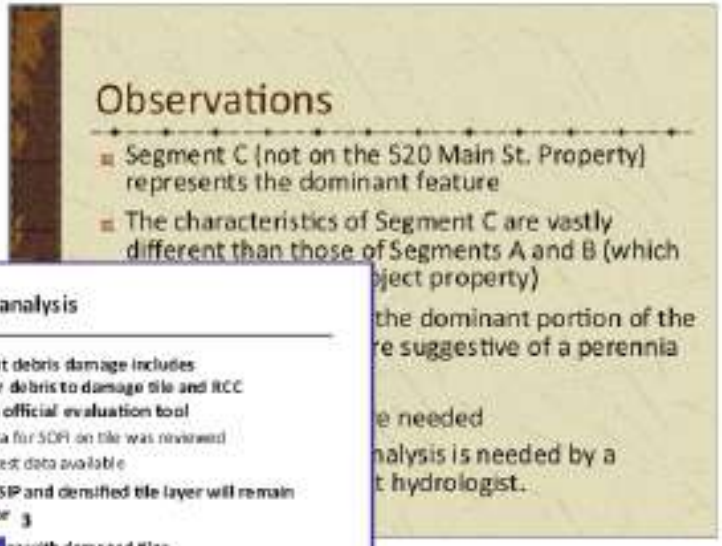
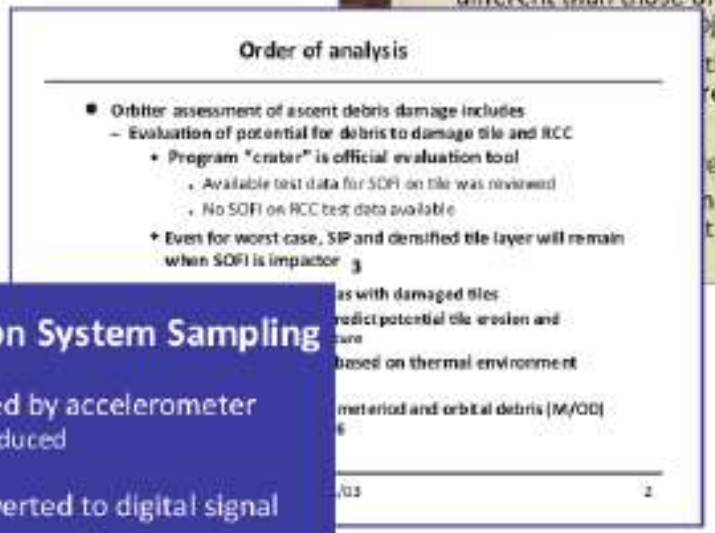
Presentation slideware's defaults lead to a topic-subtopic structure, which is ineffective for scientific presentations

In their survey, more than two-thirds of all slides

Topic-Subtopic
40%





Digital Acquisition System Sampling

- Vibration measured by accelerometer
  - Analog voltage produced
  - Sinusoidal shape
- Analog signal converted to digital signal
- Signal sampled at a specific rate
- Rate → high enough to retain analog shape
- Signal exported to popular applications

Common practice– we can do better

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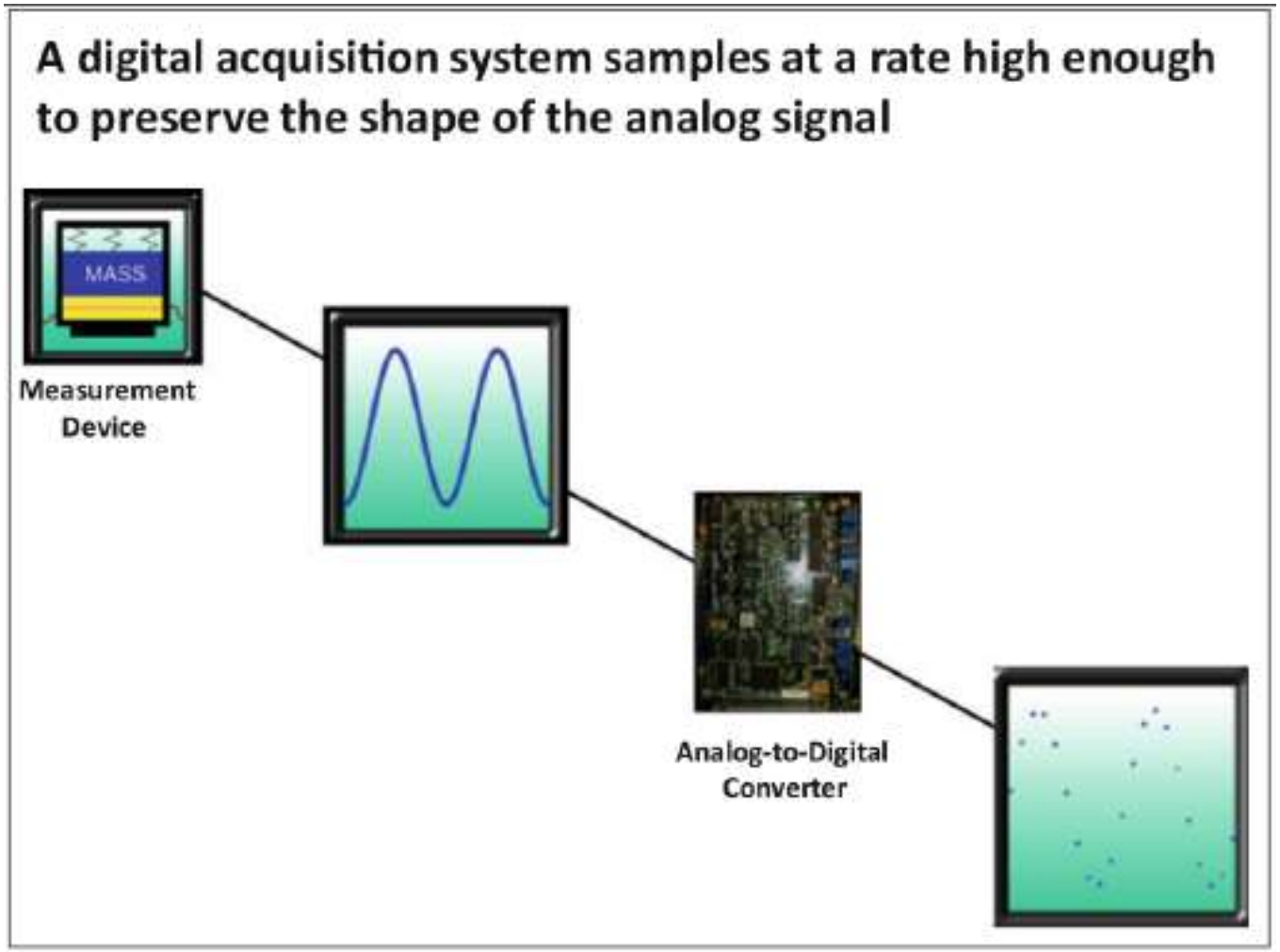
Common practice? – **we can do better**



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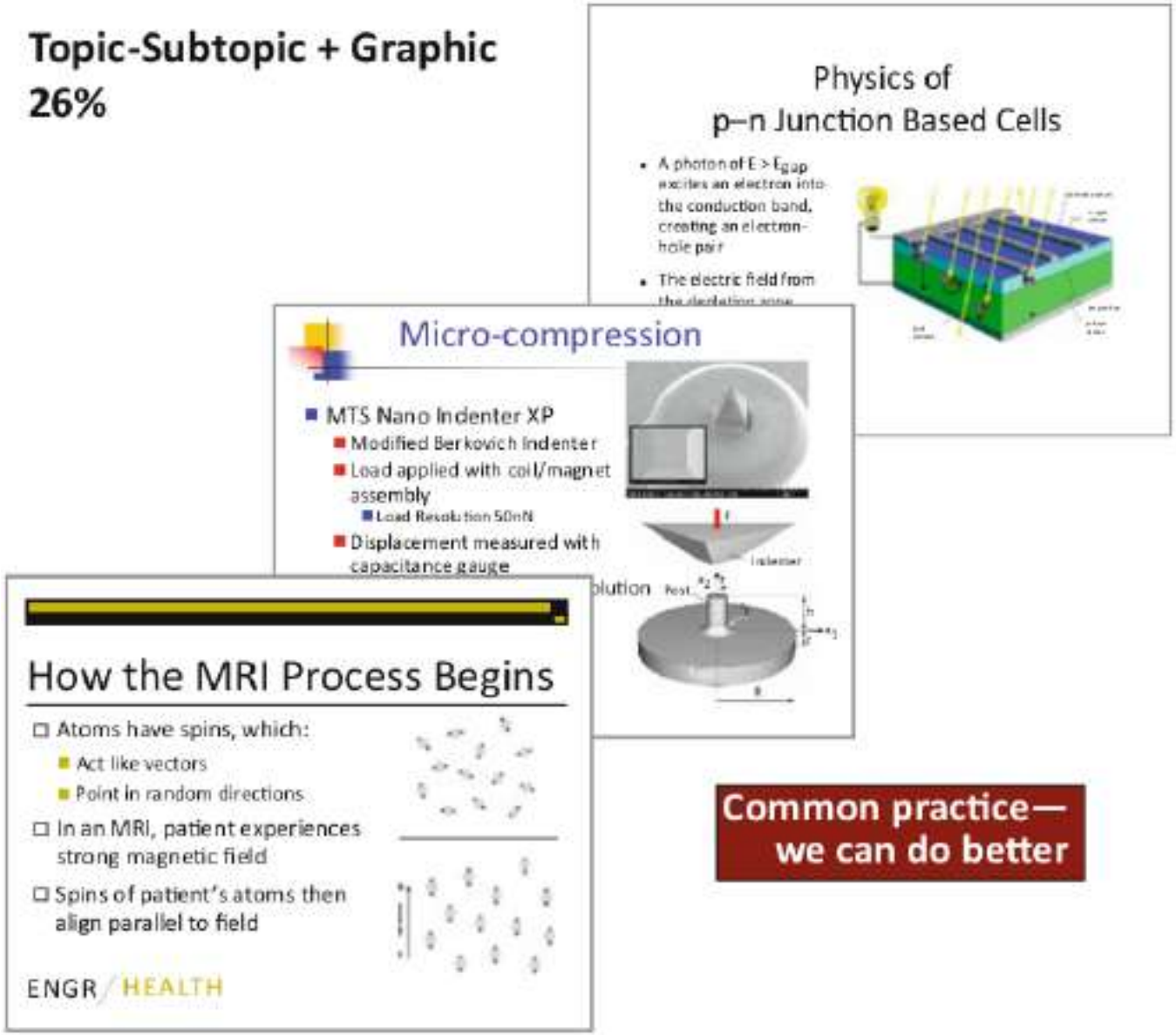
In their survey, more than a quarter of all slides

Topic-Subtopic + Graphic
26%



Common practice—we can do better

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Three ways that presentation slides affect the success: **preparation, delivery, and audience understanding**

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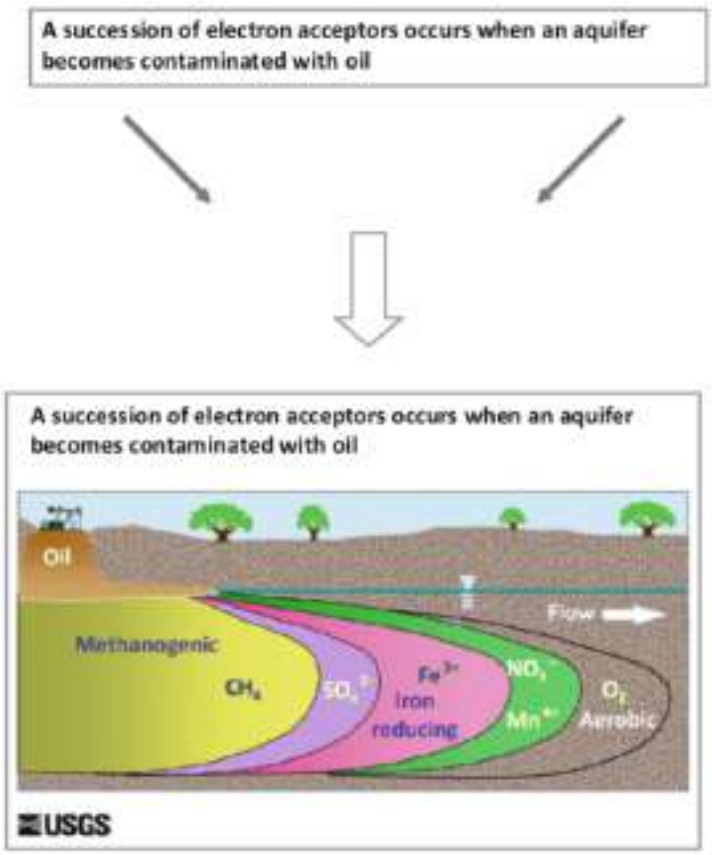
Assertion-evidence slides lead to more focused talks and more engaging deliveries



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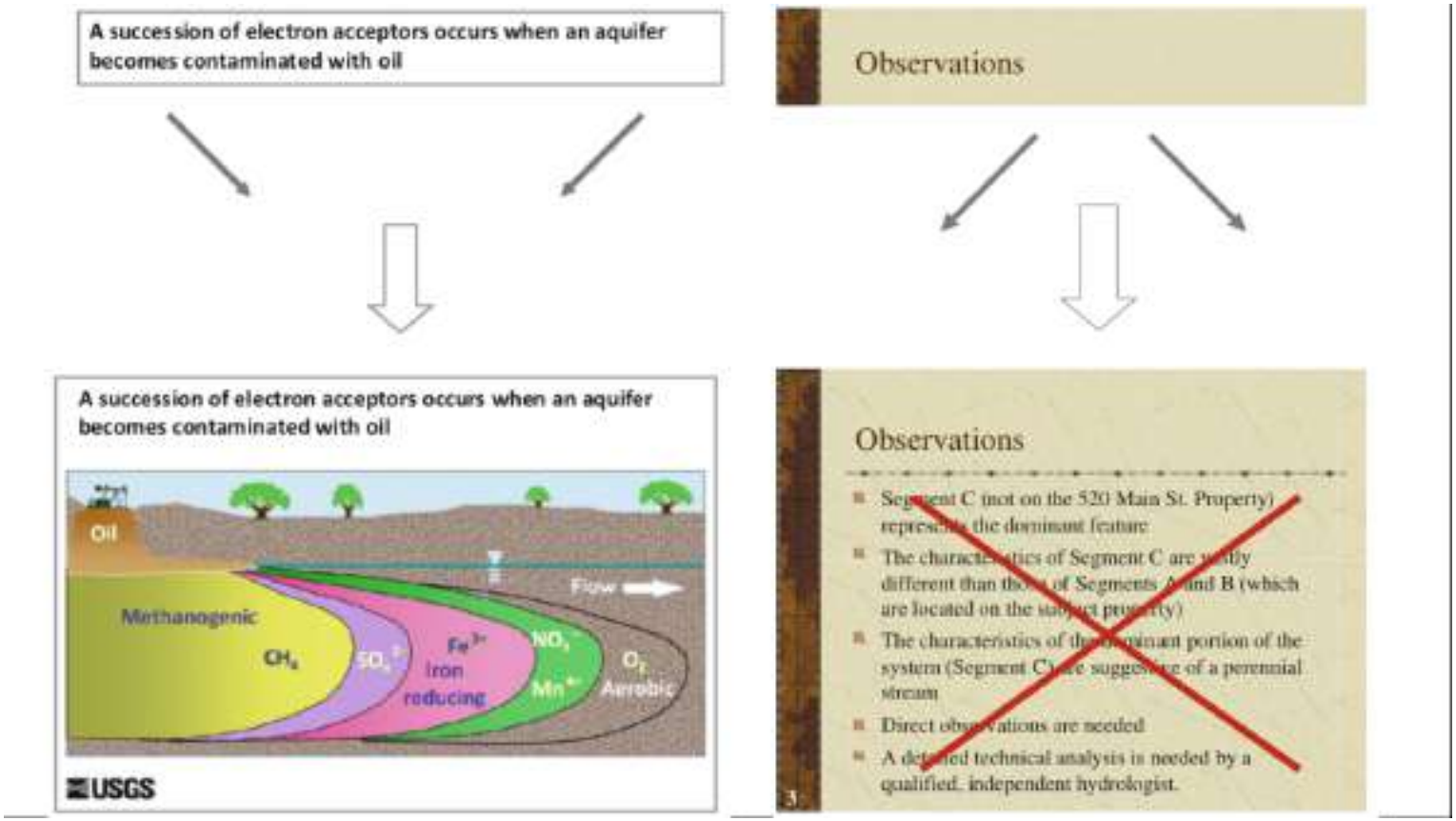
Assertion-evidence slides lead to more focused talks and more engaging deliveries



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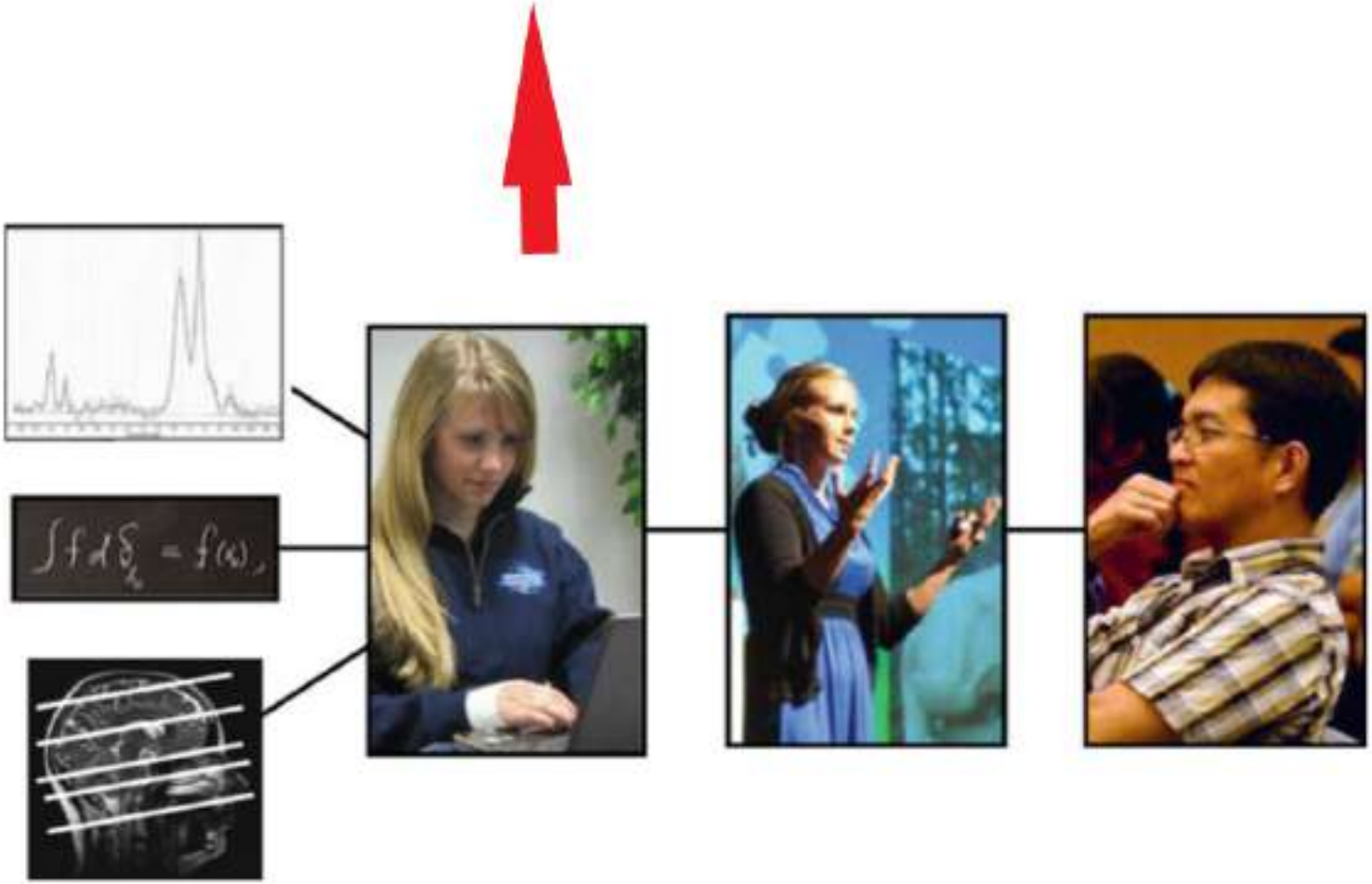
Critical Error 7: Following the Defaults of electronic presentations

- In **Critical Error 7: Following the Defaults of electronic presentations**, Trying to Cover Too Muchyou assess what
- The **CRITICAL ERROR 7: Following the Defaults of electronic presentations**
- **CRITICAL ERROR 7**, Trying to Cover Too Muchalthough often overlooked, can greatly affect the way you present

An assertion-evidence slide calls for a succinct **sentence headline** that states the slide's main assertion

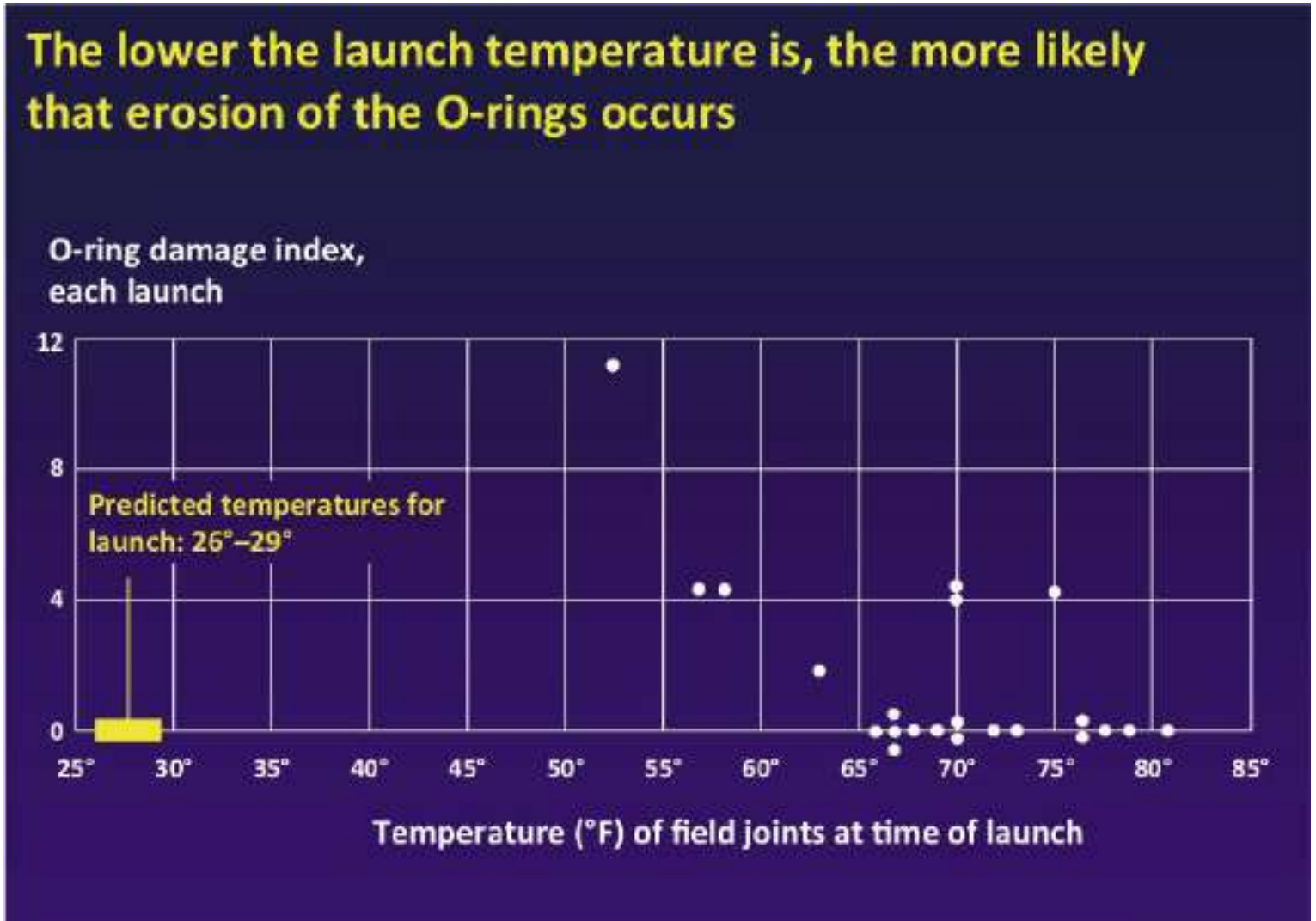
headlines help audience and speaker



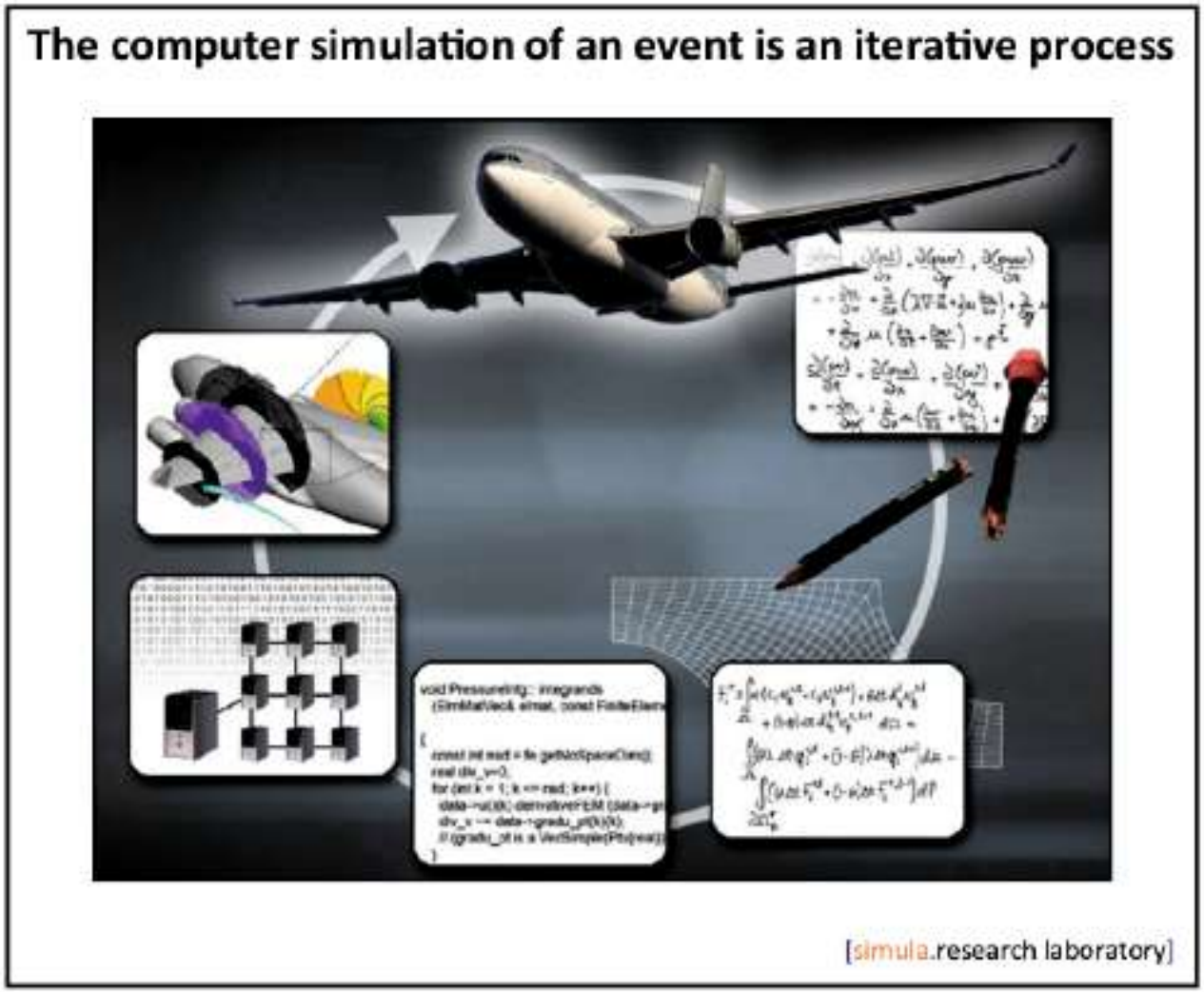
ME-USP

An assertion-evidence slide calls for supporting the headline with visual evidence, **not bulleted lists** main assertion

headlines help audience and speaker

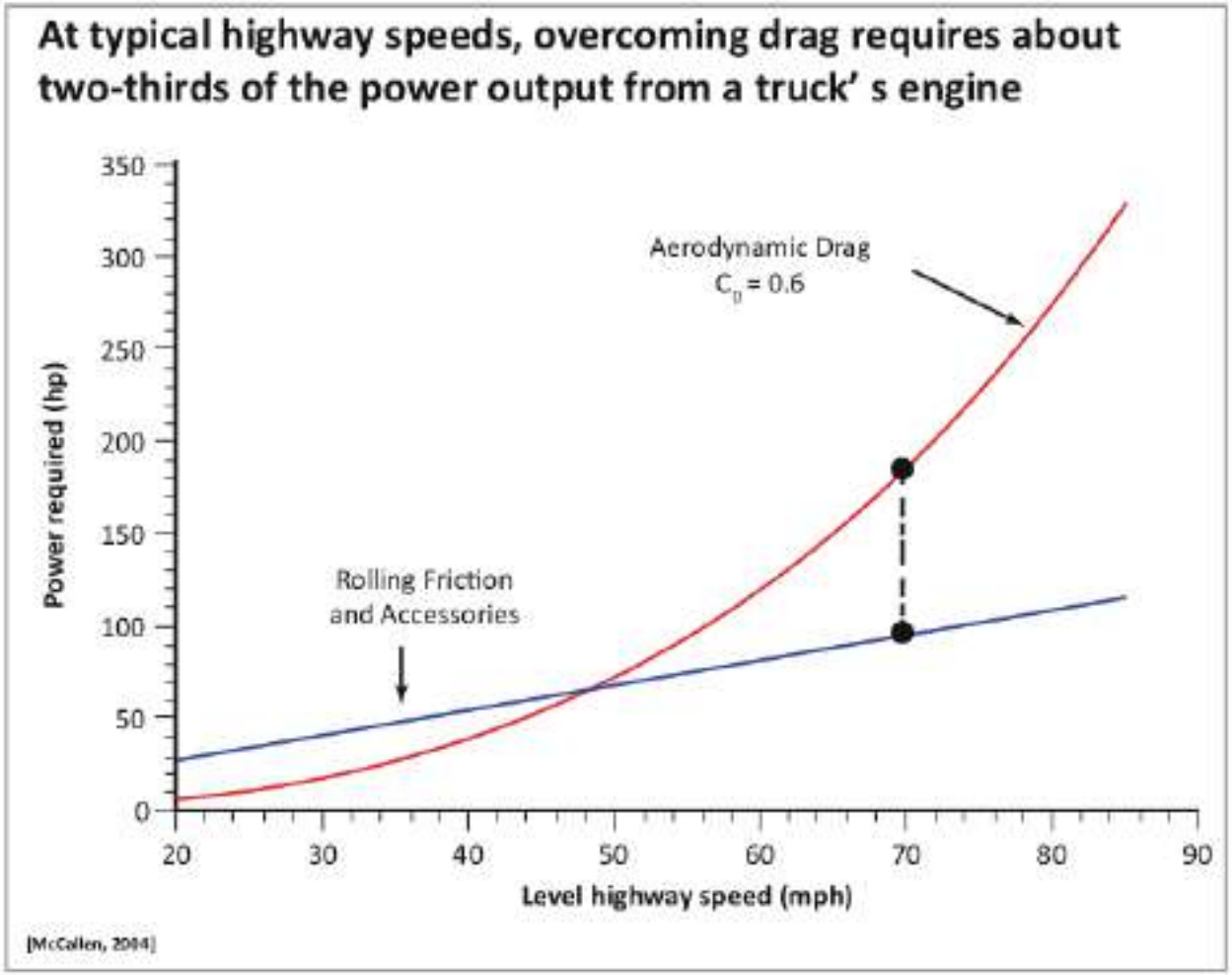


An assertion-evidence slide calls for supporting the headline with visual evidence, **not bulleted lists** main assertion



IME-USP

An assertion-evidence slide calls for supporting the headline with visual evidence, **not bulleted lists** main assertion



IME-USP

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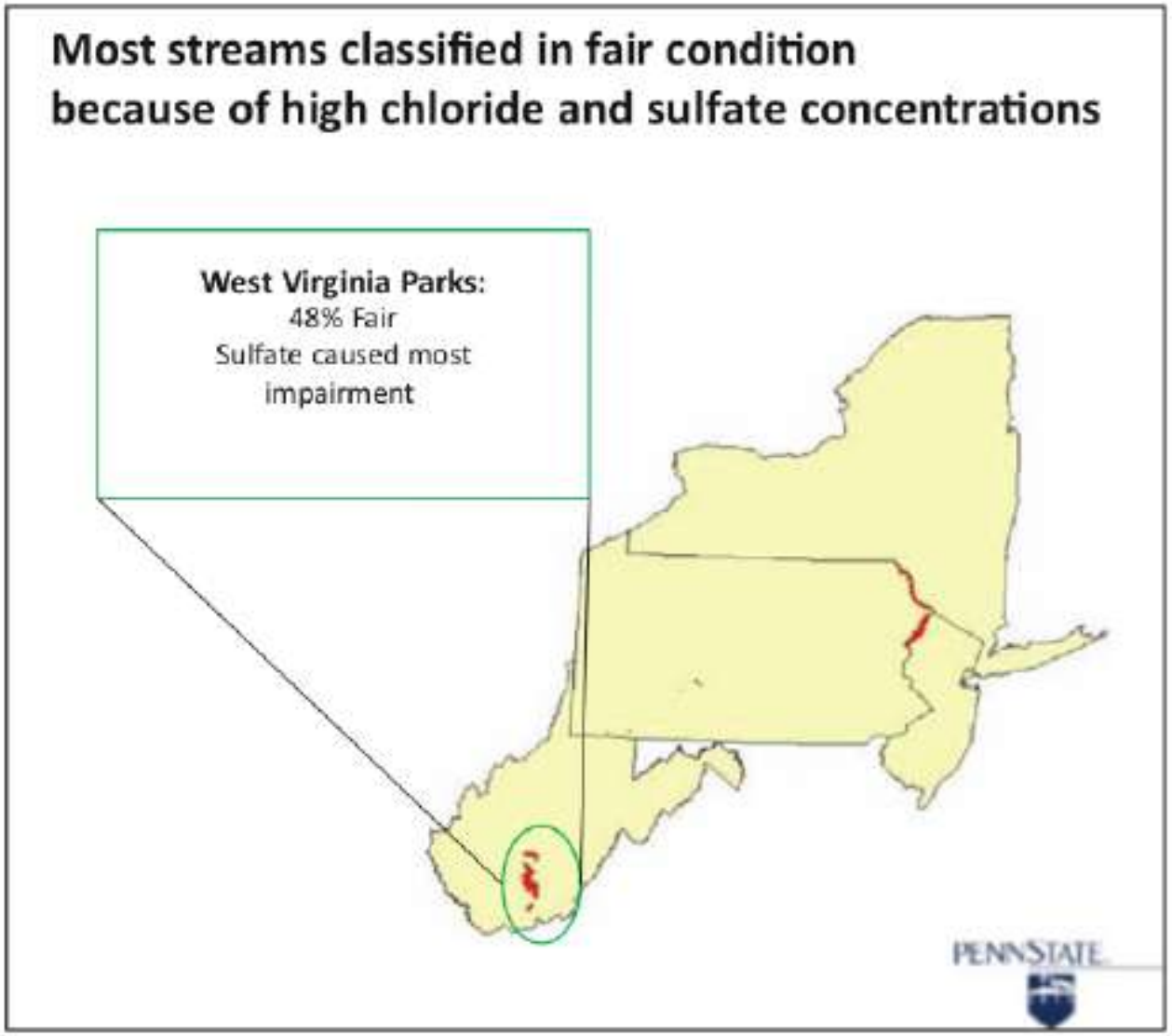

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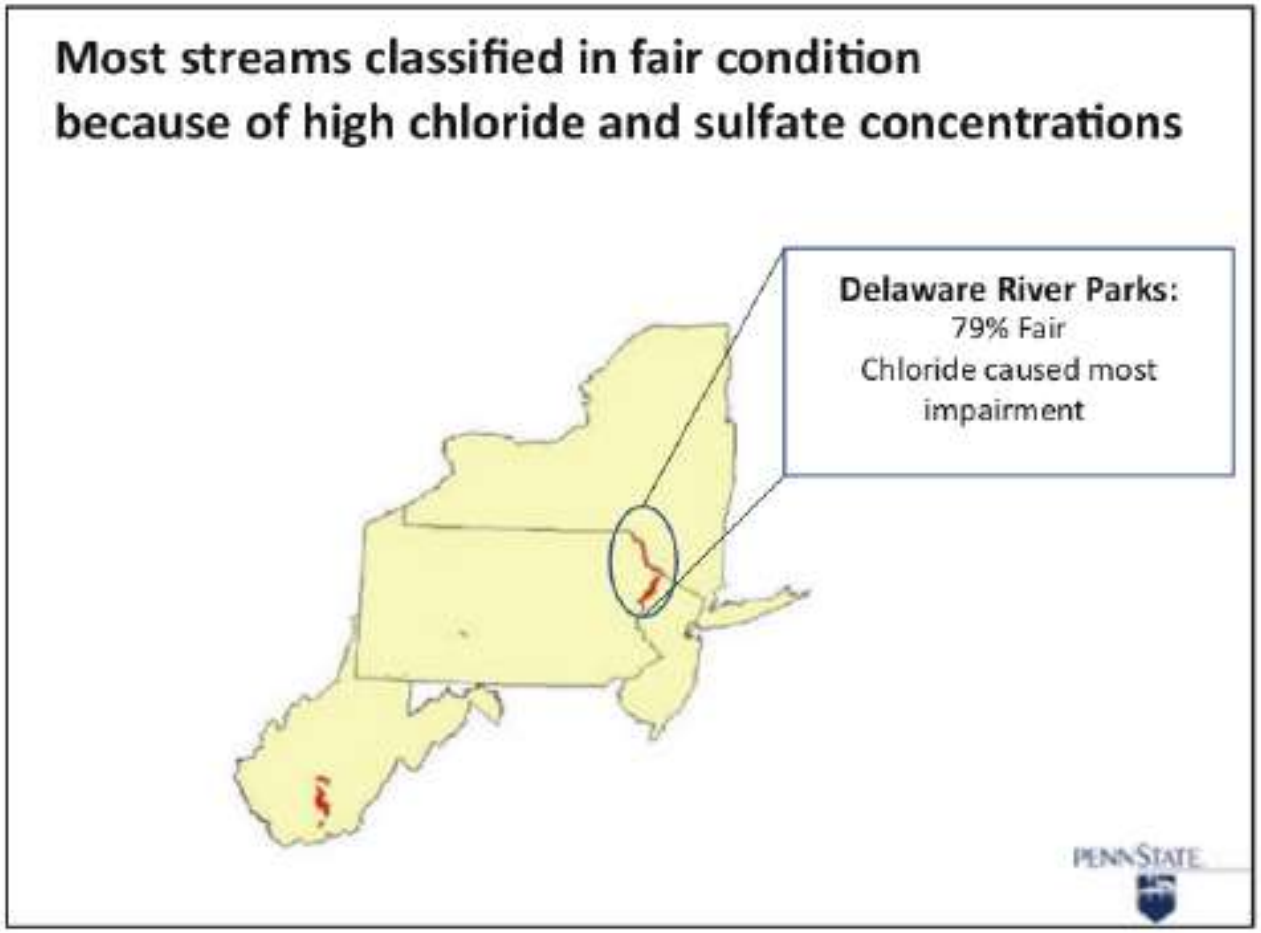

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For slides to be effective, the **format must rise above** Slideware's defaults

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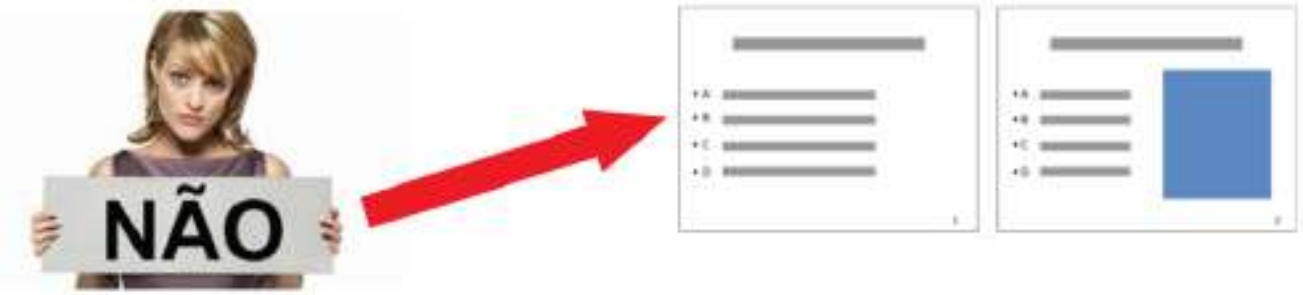
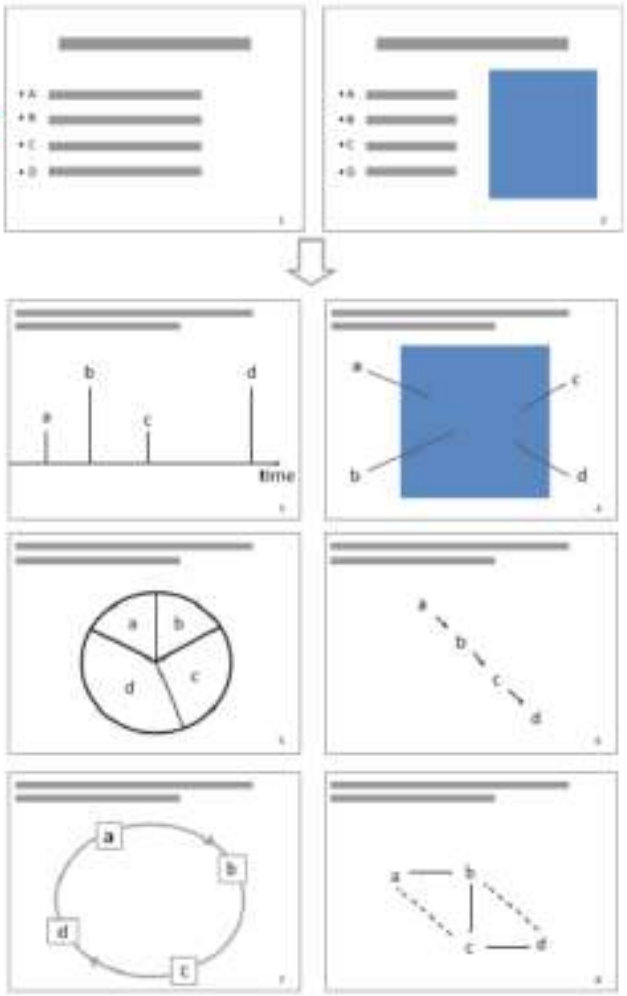


For slides to be effective, the **format must rise above** Slideware's defaults



For slides to be effective, the **format must rise above** Slideware's defaults

28 points	Headline of slide
24 points	Supporting text for body of slide
20 points	Supporting text for body of slide
18 points	Supporting text for body of slide
14 points	Reference listing (not bold)
12 points	Reference listing, small room (not bold)

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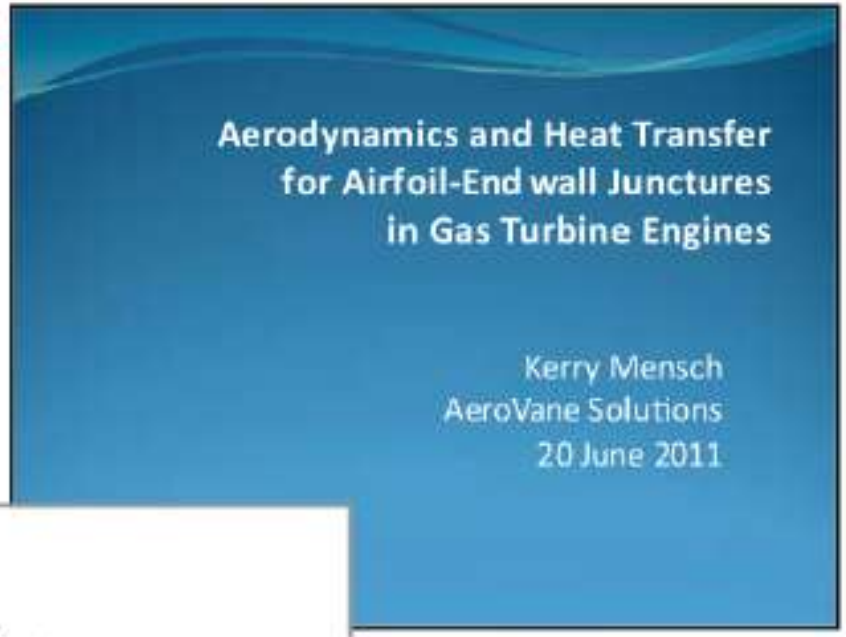
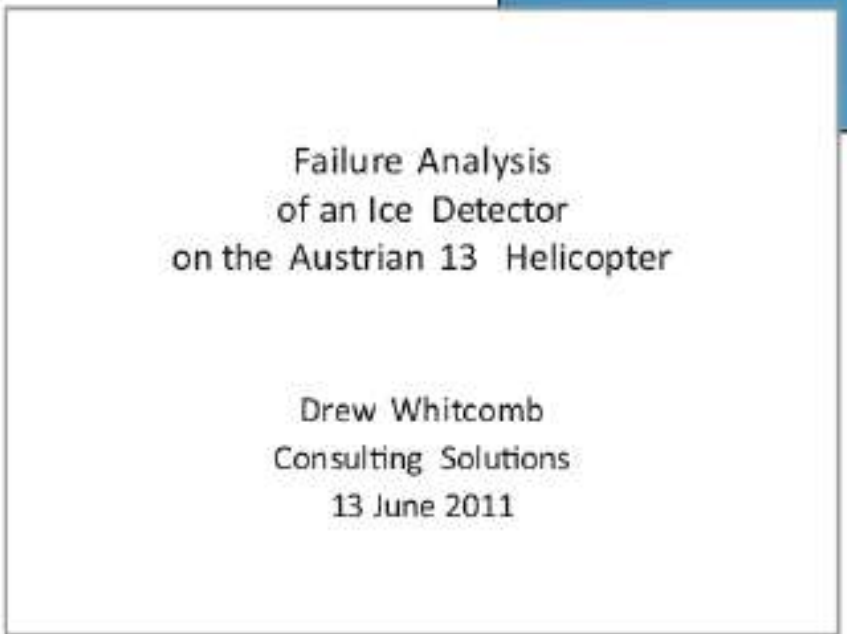
Critical Error 8: Following the Common Practices of Electronic Presentations Talks

- In **Critical Error 8: Following the Common Practices of Electronic Presentations Talks**, Trying to Cover Too Muchyou assess what they know, why
- The **Critical Error 8: Following the Common Practices of Electronic Presentations Talks** informing and persuading — and sometimes inspiring
- **Critical Error 8: Following the Common Practices of Electronic Presentations Talks**,

Title slides should orient, outline slides should map, and concluding slides should emphasize

Weak title slides:
We can do better





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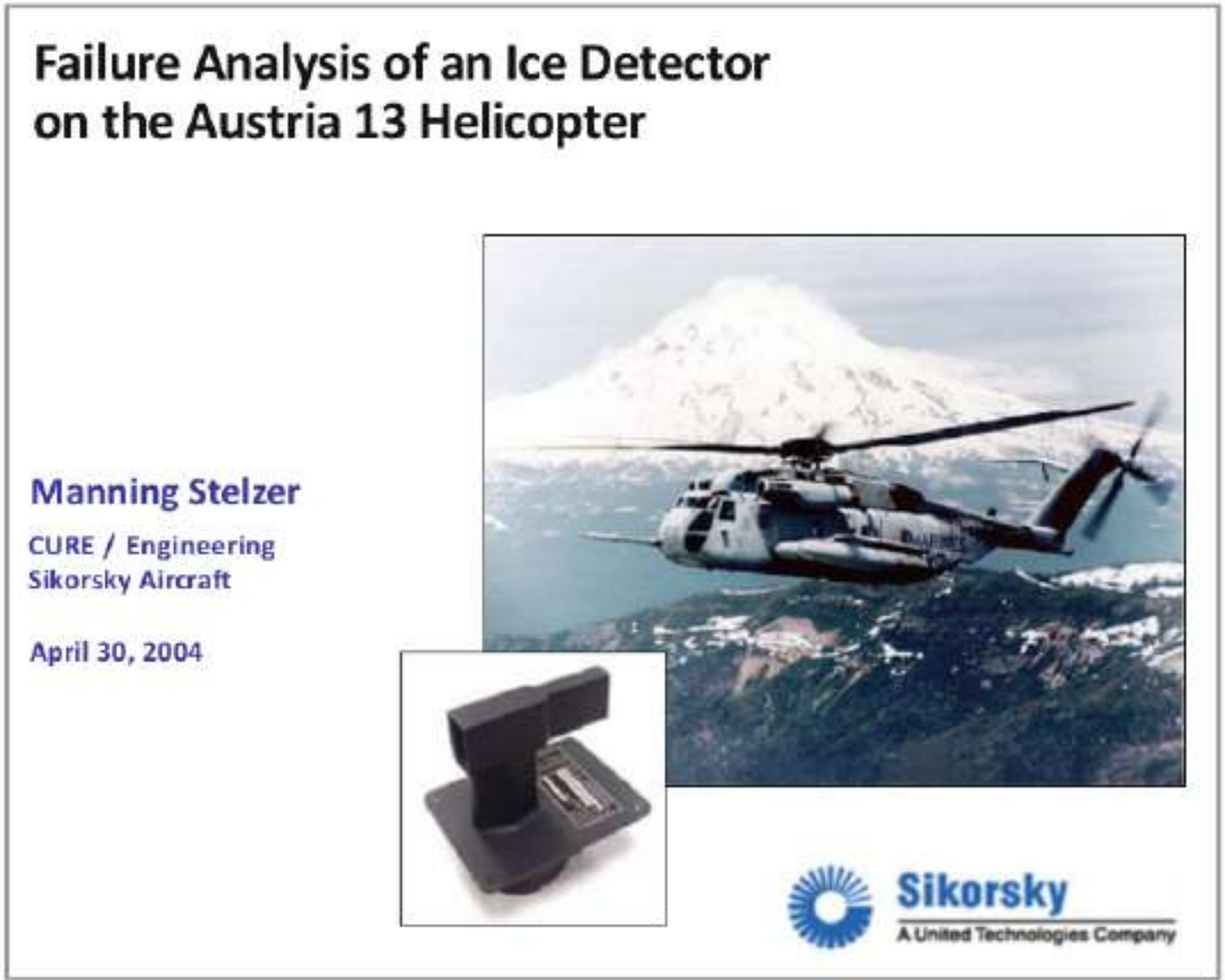
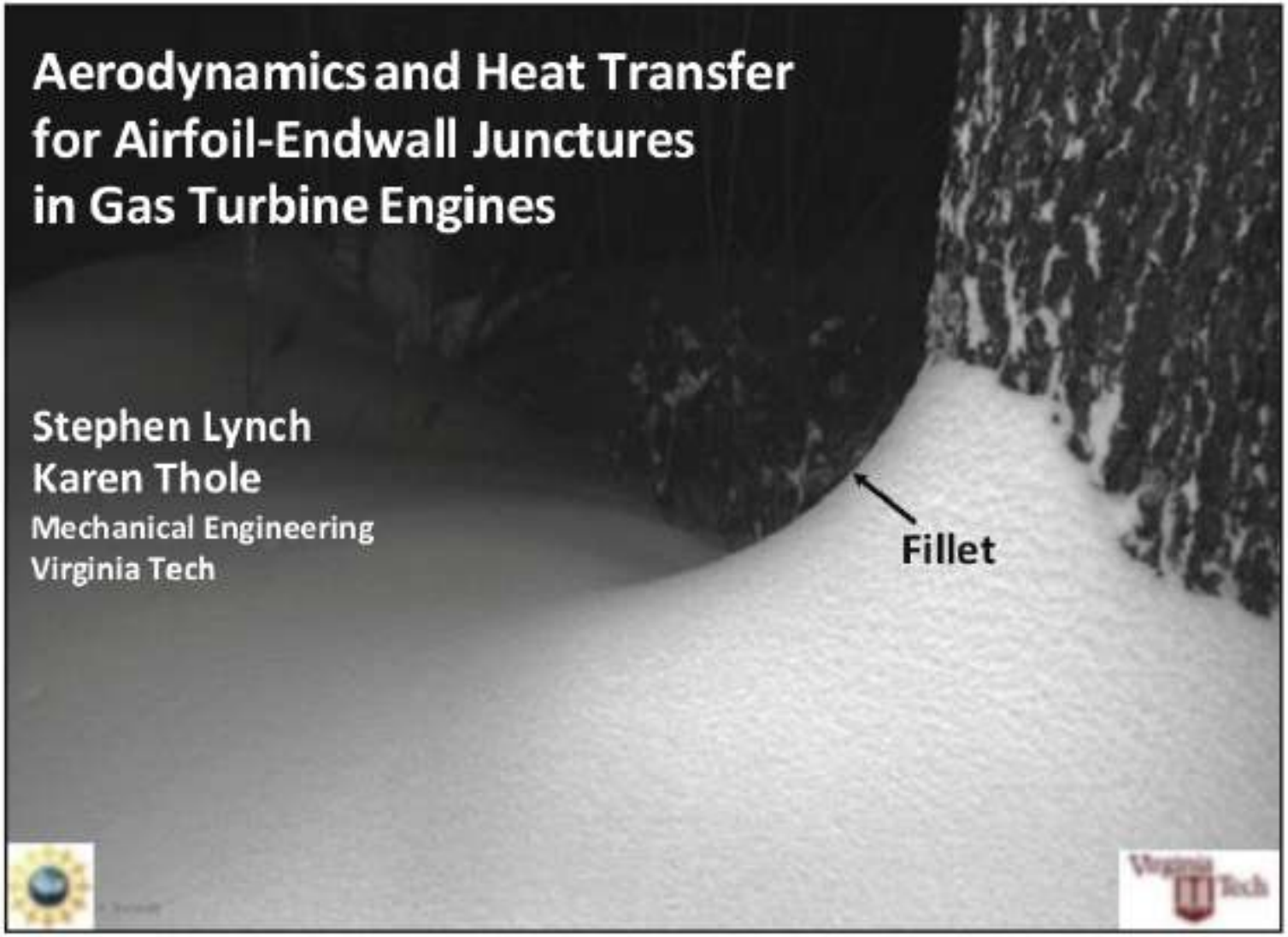
Title slides should orient, outline slides should map, and concluding slides should emphasize



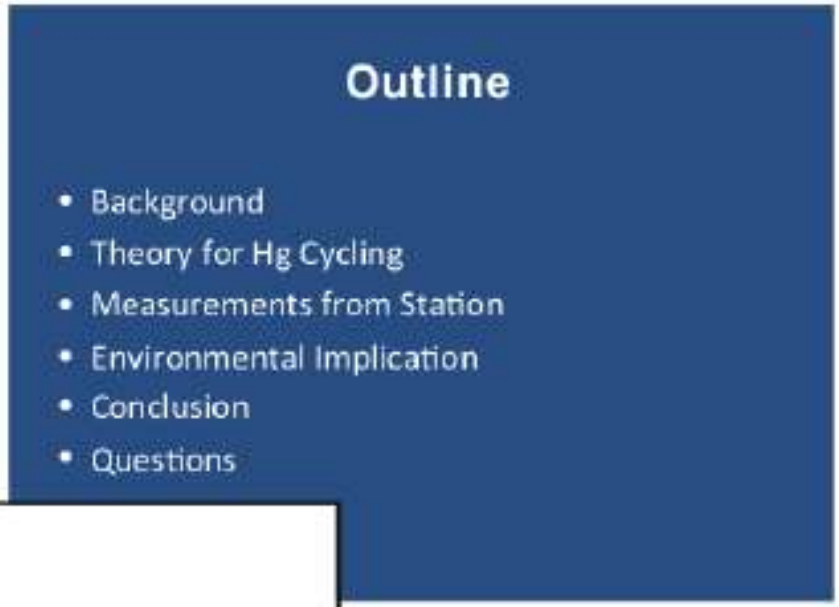
Title slides should orient, outline slides should map, and concluding slides should emphasize



IME-USP

Title slides should orient, **outline slides should map**, and concluding slides should emphasize

Weak mapping slides:
We can do better



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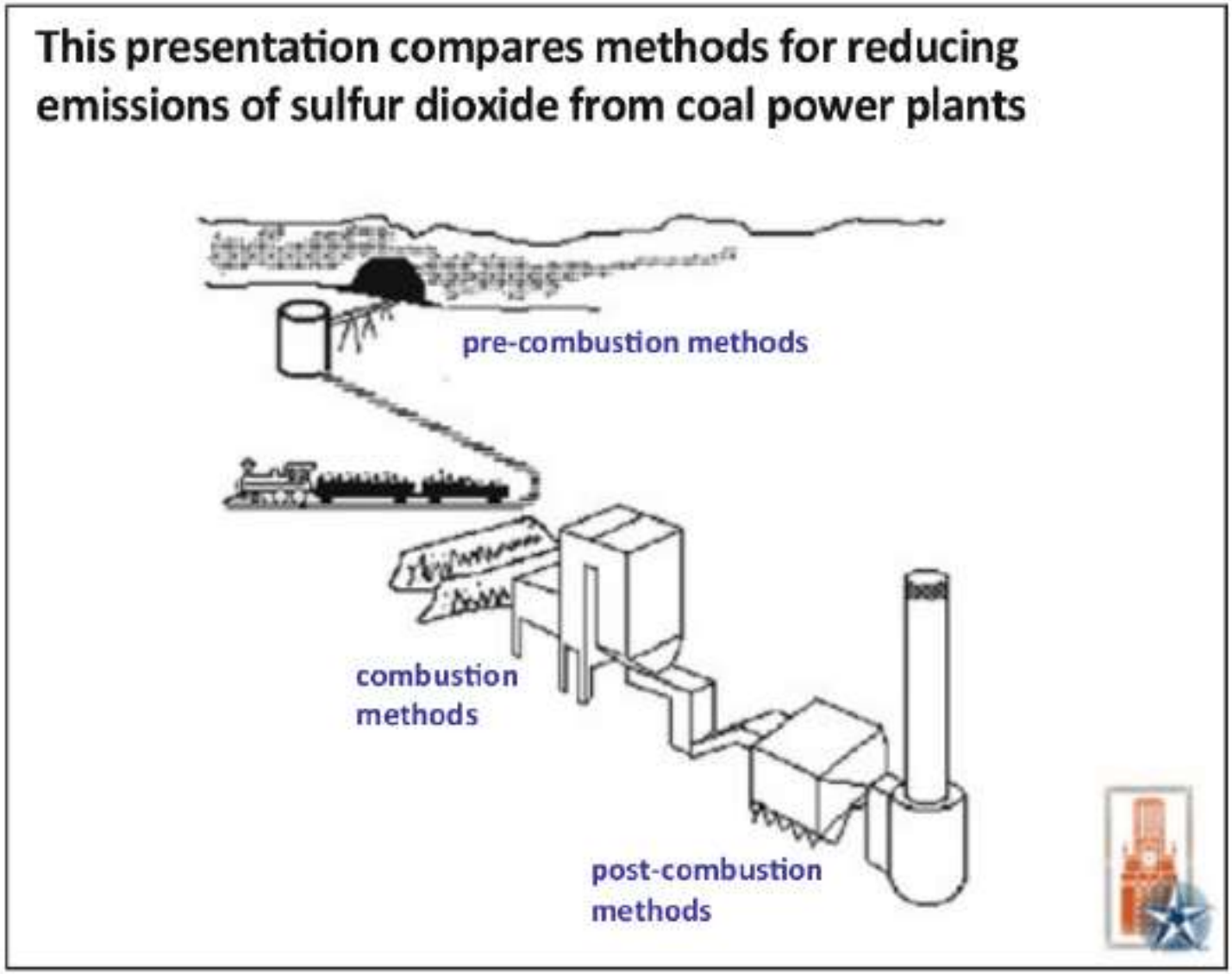
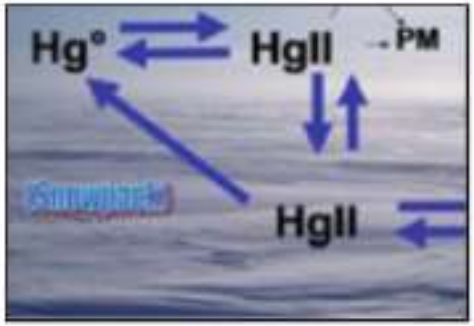
Title slides should orient, **outline slides should map**, and concluding slides should emphasize



Title slides should orient, **outline slides should map**, and concluding slides should emphasize

This talk compares the theory for mercury depletion with our measurements and gives environmental implications



Theory for Hg depletion

Measurements from station

Environmental implications

Title slides should orient, **outline slides should map**, and concluding slides should emphasize

Operator splitting gives a sequence of PDEs and one ODE per spatial point

$$\frac{\partial c_i(x,t)}{\partial t} = -\frac{\partial}{\partial x}[f_i(x,t)] \quad \text{for } x \in \Omega$$

Concentration c_i of specie #i, i=1,...,n_{ion}

$$f_i(x,t) = -D_i(x,t)\frac{\partial c_i(x,t)}{\partial x} - k_i(x,t)c_i(x,t),$$

$$k_i(x,t) = -D_i(x,t)z_i\frac{F}{RT}E(x,t)$$

Unknowns c_i and E

$$\varepsilon\frac{\partial E(x,t)}{\partial t} = I - F\sum_{i=1}^{n_{ion}} z_i f_i(x,t) \quad \text{for } x \in \Omega$$

Electrical field E

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Title slides should orient, outline slides should map, and **concluding slides** should emphasize

Weak concluding slides:
We can do better

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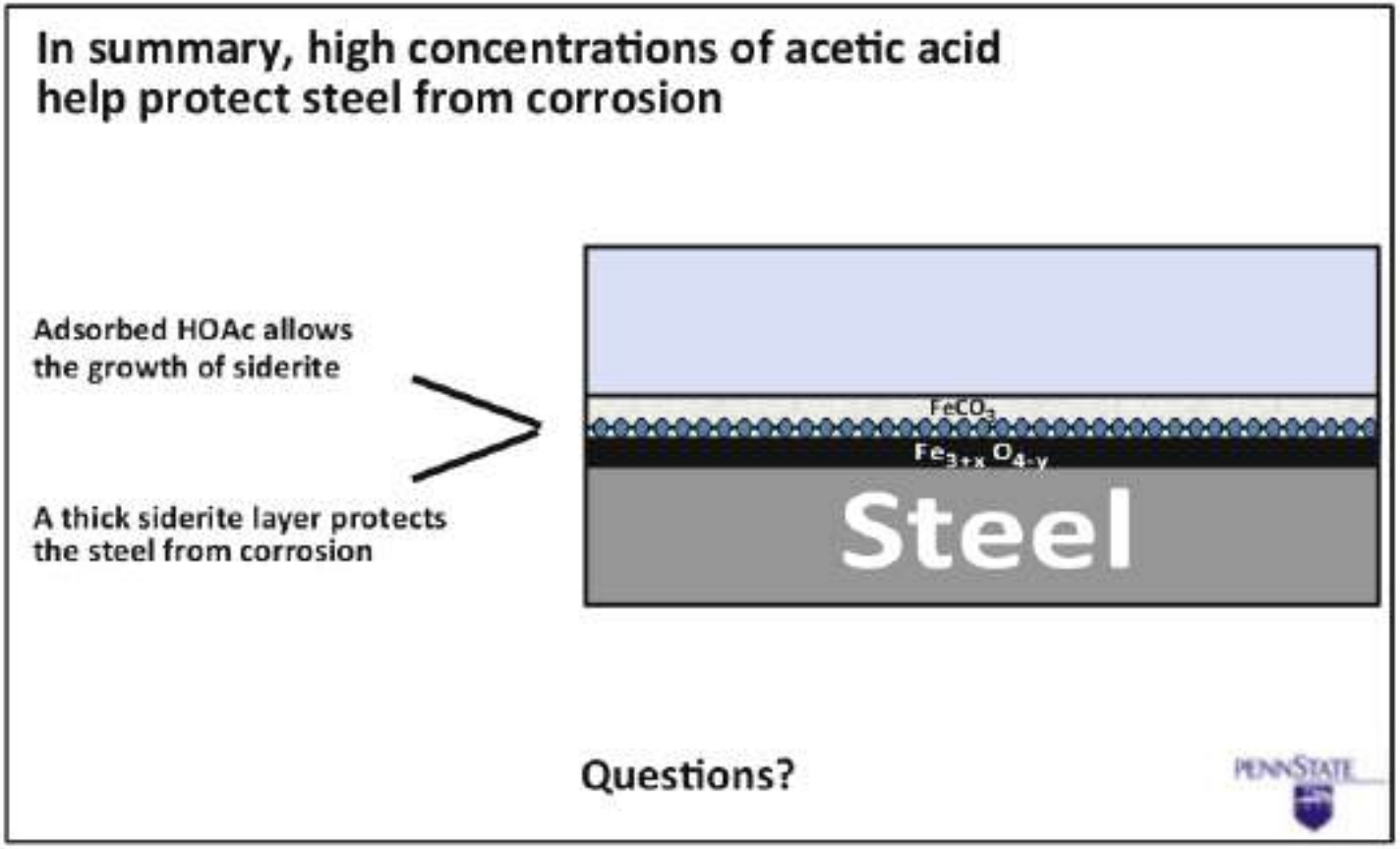
Title slides should orient, outline slides should map, and **concluding slides** should emphasize



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Title slides should orient, outline slides should map, and **concluding slides** should emphasize

¿Preguntas?

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The TED slide structure

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The TED slide structure

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Sequencing slides can be effective by:

- An evidence-assertion order,
- Pecha kucha,
- The Lessig style, and
- Prezi

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Critical Error 9: Not Accounting for Murphy's Law

- In **Critical Error 9: Not Accounting for Murphy's Law**
- The **Critical Error 9: Not Accounting for Murphy's Law**
- **Critical Error 9: Not Accounting for Murphy's Law**

Critical Error 9: Not Accounting for Murphy's Law

- **Weigh the risks** of incorporating visual aids
- Should **rehearse**
- You should **arrive early** to the room
- With presentations, you should **prepare for the worst**

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Summary

1. Introduction
2. Speech: The Words You Say
3. Structure: The Strategy You Choose
4. Visual Aids: Your Supporting Cast
5. Delivery: You, the Room, and the Audience
6. Conclusion

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The appropriate delivery depends on the speaker and the situation

You can significantly improve your delivery with practice and reflection

Critical Error 10: Not Preparing Enough

- Critical Error 10: Not Preparing Enough
- Critical Error 10: Not Preparing Enough
- Critical Error 10: Not Preparing Enough

Critical Error 10: Not Preparing Enough

1. Before opening the computer,

Critical Error 10: Not Preparing Enough

1. Before opening the computer,
2. **You should decide upon the story of the talk**

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Critical Error 10: Not Preparing Enough

1. Before opening the computer,
2. **You should decide upon the story of the talk**
3. You are **in position to create your visual aids**

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Critical Error 10: Not Preparing Enough

1. Before opening the computer,
2. **You should decide upon the story of the talk**
3. You are **in position to create your visual aids**
4. In addition to preparing **yourself to speak**

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Critical Error 10: Not Preparing Enough

1. Before opening the computer,
2. **You should decide upon the story of the talk**
3. You are **in position to create your visual aids**
4. In addition to preparing **yourself to speak**
5. Speaking in a **second language** requires additional preparation

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Critical Error 11: Drawing Words from the Wrong Well

- Critical Error 11: Drawing Words from the Wrong Well,
- Critical Error 11: Drawing Words from the Wrong Well
- Critical Error 11: Drawing Words from the Wrong Well

Critical Error 11: Drawing Words from the Wrong Well

- For most scientific presentations, a **practiced extemporaneous talk** is the best overall strategy
- While not the approach to choose for a planned talk, **impromptu speaking** is an important skill
- **Memorizing** can be effective for **short portions of talks** such as first and last sentences
- **Reading** is sometimes necessary when the audience will **scrutinize** your every word

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Critical Error 12: Not Paying Attention

- Critical Error 12: Not Paying Attention,
- Critical Error 12: Not Paying Attention
- Critical Error 12: Not Paying Attention

Critical Error 12: Not Paying Attention

- Pay attention to the room
- Pay attention to yourself
- Pay attention to the audience
- Pay attention to the time

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Critical Error 13: Losing Composure

- Critical Error 13: Losing Composure,
- Critical Error 13: Losing Composure
- Critical Error 13: Losing Composure

Critical Error 13: Losing Composure

- Preparing, thinking positively, and having the proper focus are keys to handling nervousness
- In handling a question, you should understand what was asked, think about that question, and answer honestly
- Passion fuels preparation, which leads to confidence

Summary

1. Introduction
2. Speech: The Words You Say
3. Structure: The Strategy You Choose
4. Visual Aids: Your Supporting Cast
5. Delivery: You, the Room, and the Audience
6. Conclusion

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Physicists: Story 3

https://elikatakimoto.com/2013/06/08/dialogos/

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10 tips to develop confidence in conference presentations

- Expect to be nervous
- Prepare
- Practice
- Breathe
- Rehearse
- Focus on your audience
- Simplify
- Picture success
- Connect with your audience
- Be confident

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Scientific Presentation (Seminary)

¿Questions?

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Scientific Presentation (Seminary)

Luiz Henrique Rodrigues

InterSCity Project
Software Sistems Lab
IME - USP

May 10, 2019

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